Combinatorics 2019 Fall

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1 SDR, Latin Rectangle, Permutation Matrix

Theorem 1.1. *(**Hall's marriage Theorem**) The sets of $S_1, S_2, \cdots, S_m$ has a SDR $\Longleftrightarrow$ $|\cup_{i\in I} S_i| \geq |I|$ for $I \subset [m]$(Hall's condition).*

Corollary 1.2. *$|X| = n$, $|S_i| = r, S_i \subset X, i \in [m].s.t.|\{i : x \in S_i\}| = d$ for all $x \in X$. If $m \leq n$, then $S_1, \cdots, S_m$ have an SDR.*

Theorem 1.3. *Suppose elements in X are colored by either red or blue. $S_i \subset X, i \in [m]$, then $S_1, \cdots, S_m$ have a SDR with $\leq t$ red elements iff $S_1, \cdots, S_m$ have a SDR and $\forall I \subset [m], \cup_{i\in I} S_i$ has $\geq |I| - t$ blue elements.*

Proof. "$\Longrightarrow$" Let $x_1, \cdots, x_m$ be a SDR of $S_1, \cdots, S_m$ with $\leq t$ red elements. then $\forall I \subset [m]$, $\{x_i, i \in I\}$ has at least $|I| - t$ blue elements $\Rightarrow \cup_{i\in I} S_i$ has at least $|I| - t$ blue elements.

"$\Longleftarrow$" Let R be the set of red elements in X. If $|R| \leq t$, trivial. Assume $|R| > t$, let $S_{m+1} = S_{m+2} = \cdots = S_{m+r} = R$, where $r = |R| - t$. then $S_1, \cdots, S_m$ have a SDR with $\leq t$ red elements $\Longleftrightarrow$ $S_1, \cdots, S_m, S_{m+1}, \cdots, S_{m+r}$ have a SDR. So we need to check Hall's conditio n for $S_1, \cdots, S_{m+r}$, let $Y = \cup_{i\in I} S_i$, if $I \subset [m]$, then $|Y| \geq |I|$ since $S_1, \cdots, S_m$ have a SDR. if $I = J_1 \cup J_2$, where $J_1 \subset [m], J_2 \subset [m+1, m+r]$, then $|J_2| \leq |R| - t, |Y| = |\cup_{i\in J_1} (S_i \backslash R)| + |R| \geq |J_1| - t + |R| = |J_1| + (|R| - t) \geq |J_1| + |J_2| = |I|$. $\square$

Application

Definition. *A $r \times n (r \leq n)$ Latin rectangle is $r \times n$ matrix over $[n]$ s.t.numbers $1, 2, \cdots, n$ occurs once in each row and $\leq$once in each column. A Latin square is an $n \times n$ Latin rectangle.*

Theorem 1.4. *(**Evans conjecture**) If fewer than n cells of an $n \times n$ matrix are filled, then one can always complete it into a Latin square.*

Theorem 1.5. *If $r < n$, then any given $r \times n$ Latin rectangle can be extended to an $(r + 1) \times n$ Latin rectangle.*

Proof. Let R be $r \times n$ LR, For $j \in [n]$, let S_j be the set of integers in $[n]$ which don't occur in the j-th column. Then it suffices to prove $S_1, \cdots, S_n$ have a SDR. Since $|S_j| = n - r$, and each $i \in [n], i$ occurs in $n - r$ sets S_j, by Corollary1, $S_1, \cdots, S_n$ have a SDR. □

Definition. *An $n \times n$ matrix $A = \{A_{ij}\}$ with $a_{ij} \geq 0$ is called doubly stochastic if $\sum_{j=1}^{n} a_{ij} = \sum_{i=1}^{n} a_{ij} = 1$ for $\forall i, j \in [n]$. If $a_{ij} = 0$ or 1, then it is a permutation matrix.*

Theorem 1.6 (Birkhoff)**.** *Every doubly stochastic matrix A is a convex combination of permutation matrixes, that is, $\exists$ permutation matrixes $P_1, \cdots, P_s$ and non-negative reals $\lambda_1, \cdots, \lambda_s$ s.t.$A = \sum_{i=1}^{s} \lambda_i P_i$ and $\sum_{i=1}^{s} \lambda_i = 1$.*

Proof. Let A be an $n \times n$ doubly stochastic matrix, let m be the number of non-zero entries in A, then $m \geq n$. prove by induction on m. If $m = n$, then each non-zero entry is 1, so A itself is a permutation matrix. If $m > n$ and the results holds for matrices with $< m$ non-zero entries. Define $S_i = \{j : a_{ij} > 0\}, i \in [n]$. If for some of the sets $S_{i_1}, S_{i_2}, \cdots, S_{i_k}, |\cup_{i=1}^{k} S_{i_k}| \leq k - 1$. that is all non-zero entries in rows $i_1, \cdots, i_k$ occupy at most $k - 1$ columns, say columns $j_1, \cdots, j_{k-1}$, if count by rows, we have the sum is k, but if count by columns, the sum is at most $k - 1$, a contradiction. By Hall's Theorem, there is a SDR $j_1 \in S_1, j_2 \in S_2, \cdots, j_n \in S_n$. Take a permutation matrix $P_1 = (P_{ij})$ with entries $p_{ij} = 1$ iff $j = j_i$. Let $\lambda_1 = min\{a_{1j_1}, \cdots, a_{nj_n}\}$. and consider $B_1 = A - \lambda_1 P_1$. By definition of S_i, we have $\lambda_1 > 0$, matrix B_1 has at most $m - 1$ non-zero entries, and the row sum and column sum of B_1 is $1 - \lambda_1$. Let $A_1 = \frac{1}{1-\lambda_1} B_1$, then A_1 is a doubly stochastic matrix with less than m non-zero entries. By assumption $A_1 = \mu_2 P_2 + \cdots + \mu_s P_s$ a convex combination. Hence, $A = \lambda_1 P_1 + (1 - \lambda_1) A_1 = \lambda_1 P_1 + (1 - \lambda_1)\mu_2 P_2 + \cdots + (1 - \lambda_1)\mu_s P_s$. Since $\sum_{i=2}^{s} \mu_i = 1$, we have $\lambda_1 + (1 - \lambda_1)(\sum \mu_i) = 1$.

□

Definition. *Let $S_1, \cdots, S_m \subseteq X = \{x_1, \cdots, x_n\}$, and $M = (a_{ij})$ be the corresponding incidence matrix. The* ***permanent*** *of M is*

$$Per(M) = \sum_{(i_1, \cdots, i_m) \in S_n(m)} a_{i_1 1} a_{i_2 2} \cdots a_{i_m m},$$

where $S_n(m)$ is the set of all vectors of length m over $[n]$ without repetition.

Fact: $Per(M) = \sharp$different SDR's of $S_1, \cdots, S_m$.

Definition. *Let A be a 0-1 matrix. Two 1's are dependent if they are in the same row or the same column, otherwise, they are independent.*

Theorem 1.7. *(***König***) Let A be an $m \times n$ 0-1 matrix, then max $\sharp$ independent 1's r = the min $\sharp$ rows and columns R required to cover all 1's in A.*

Proof. Clearly, $R \geq r$, since we can find r independent 1's and every row or column covers at most one of them.

Now we show $r \geq R$. Assume that some a rows and b columns cover all 1's and $a+b=R$. We may assume the first a rows and the first b columns cover all the 1's. Write A as the form

$$A = \begin{pmatrix} B_{a\times b} & C_{a\times(n-b)} \\ D_{(m-a)\times b} & E_{(m-a)\times(n-b)} \end{pmatrix},$$

with no 1 in $E_{(m-a)\times(n-b)}$. If we can show that there are a independent 1's in C and b independent 1's in D, then we find at least $a+b$ independent 1's, so we have $r \geq a+b=R$.

For each $1 \leq i \leq a$, let $S_i = \{j : \ c_{ij} = 1\} \subseteq [n-b]$. If $S_1, \cdots, S_a$ have an SDR, then we find a independent 1's in C If not, by Hall's theorem, there are some $k \in [a]$ sets, say $S_{i_1}, \cdots, S_{i_k}$, such that $\left| \bigcup_{j=1}^{k} S_{i_j} \right| < k$, i.e. the 1's in these k rows occupy at most $k-1$ columns of C, say $j_1, \cdots, j_{k-1}$. Then the first b columns of A, the columns $j_1, \cdots, j_{k-1}$ of C and the first a rows of A deleting the rows $i_1, \cdots, i_k$ will cover all 1's in A. So we find $b+(k-1)+a-k = a+b-1$ rows and columns cover all 1's in A, contradiction!

□

2 König's Theorem, SCR, bipartite graph, Pigeonhole Principle

Theorem 2.1 (König's Theorem)**.** *Let A be an $m \times n$ 0-1 matrix. Let r be the maximum number of independent 1's, R be the minimum number of rows and columns required to cover all 1's in A. Then, $r = R$. ($\Longleftrightarrow$ A doesn't have a 0-submatrix of size $c \times d$ such that $c+d = m+n-r+1$.)*

Definition. *A system of common representatives (SCR) of two sequences of sets $A_1, \cdots, A_m$ and $B_1, \cdots, B_m$ is a sequence $x_1, \cdots, x_m$ (not necessarily distinct) such that $x_i \in A_i \cap B_{\pi(i)}, \ i \in [m]$ for some $\pi \in S_m$.*

Theorem 2.2. *Suppose X has two partitions: $X = A_1 \cup \cdots \cup A_m = B_1 \cup \cdots \cup B_m$. Then they have an SCR if and only if for $\forall\ I \subset [m]$, $\bigcup_{i\in I} A_i$ contains at most $|I|$ sets of $B_j, \ j \in [m]$.*

Proof 1. Let $Y = \{B_1, \cdots, B_m\}$. For $\forall\ i \in [m]$, let $S_i = \{B_j : B_j \cap A_i \neq \emptyset\}$. Then $\exists$ SCR $\Longleftrightarrow S_1, \cdots, S_m$ have an SDR.
So it suffices to check that $S_1, \cdots, S_m$ satisfy the Hall Condition, i.e. $\forall\ I \subset [m]$, $|\cup_{i\in I} S_i| \geqslant |I|$.
In fact, this is equivalent to: for $\forall\ I \subset [m]$, we have at least $|I|$ sets B_j such that each B_j intersect some A_i, $i \in I$. Since $A_1, \cdots, A_m$ and $B_1, \cdots, B_m$ are partitions, this is equivalent to say that $\cup_{j\in[m]\setminus I} A_j$ contains at most $m-|I|$ sets B_j, which is just the given condition. □

Proof 2. Consider the intersection matrix of the two partitions $C = (c_{ij})$, where

$$c_{ij} = \begin{cases} 1 & \text{if } A_i \cap B_j \neq \emptyset \\ 0 & \text{else} \end{cases}.$$

Then $\exists$ SCR $\iff C$ has m independent 1's $\iff C$ has no $(m-k)\times(k+1)$ 0-submatrix for any $0 \leqslant k \leqslant m-1 \iff \nexists\, m-k$ sets A_i such that their union is disjoint with $k+1$ sets $B_j \iff \nexists\, k$ sets A_i such that their union contains $k+1$ sets $B_j \iff$ The union of any k sets A_i contains at most k sets B_j. $\square$

Definition.

- *$G=(V,E)$ is called a bipartite graph, if $V = A \sqcup B$, and $i \sim j$ only if i,j are in different sets. G is called a complete bipartite graph, if G is bipartite, and every possible edge belongs to E.*
- *Two edges are disjoint, if they have no common vertex.*
- *A matching is a set of pairwise disjoint edges.*
- *A vertex is called matched (or saturated) if it is an endpoint of one of the edges in the matching. Otherwise the vertex is called free (or unmatched).*
- *If a matching matches all vertices in A, then we say it's a matching of A into B. A perfect matching is a matching of A into B when $|A| = |B|$.*
- *A vertex $x \in A$ is a neighbor of a vertex $y \in B$, if $\{x,y\} \in E$. Let S_x be the set of all neighbors of x, and call $\deg(x) := |S_x|$ the degree of x.*

Fact. $\exists$ a matching of A into B if and only if $S_x,\ x \in A$ has an SDR.

Theorem 2.3. *If G is a bipartite graph with bipartitions A, B, then G has a matching of A into B iff $\forall$ k vertices in A have at least k neighbors.*

Definition. *A vertex cover in $G = (A \cup B, E)$ is a set of vertices $S \subseteq A \cup B$ such that every edge is incident to at least one vertex in S.*

Theorem 2.4. *For bipartite graphs, the maximum size of a matching is equal to the minimum size of a vertex cover.*

Proposition 2.5. *$|X| = n$. For any $k \leqslant \frac{n-1}{2}$, it is possible to extend every k-element subset of X to a $(k+1)$-subset such that the extensions of no two sets coincide.*

Proof. Consider the bipartite graph $G = (A \cup B, E)$, where $A = \{$all k-subsets of $X\}$, $B = \{$all $(k+1)$-subsets of $X\}$, and for any $x \in A$, $y \in B$, $x \sim y$ iff $x \subset y$. Then we only need to prove G has a matching of A into B.
For each $x \in A$, $\deg(x) = n-k$. For each $y \in B$, $\deg(y) = k+1$. For $\forall\ I \subseteq A$, let $S(I) = \bigcup\limits_{x \in I} S_x$, then

$$|I|(n-k) \leqslant |S(I)|(k+1) \Longrightarrow |S(I)| \geqslant |I|\frac{n-k}{k+1} \geqslant |I|.$$

By Theorem 2, we're done. $\square$

Pigeonhole Principle and Graphs

Remark. *Here, we only consider finite simple graphs: no loops, no multiple edges.*

Theorem 2.6. *In any finite graph, the number of vertices which have odd degrees is even.*

Proof. Double counting.

For $G = (V, E)$, count # ordered pairs $(x, y) \in E$,

$$2|E| = \sum_{x \in V} \deg(x) = \sum_{x \in V,\ \deg(x) \text{ is odd}} \deg(x) + \sum_{x \in V,\ \deg(x) \text{ is even}} \deg(x).$$

If we have odd number of vertices with odd degree, then the first item is odd and the second item is even, a contradiction. □

Definition. *$G = (V, E)$ is a graph.*

- *A set of pairwise nonadjacent vertices is called an independent set. Independent number $\alpha(G)$ is the maximum size of such a set, i.e. $\alpha(G)$ is the maximum number of pairwise nonadjacent vertices of G.*
- *A coloring of V is called proper, if no two adjacent vertices have the same color. Chromatic number $\chi(G)$ is the minimum number of colors in a proper coloring.*

Proposition 2.7. *In any graph G with n vertices, $n \leqslant \alpha(G)\chi(G)$.*

Proof. Given a proper coloring of G with $\chi(G)$ colors and partition $V(G)$ into $\chi(G)$ color classes. By Pigeonhole Principle, one of the color classes has size $\geqslant \frac{n}{\chi(G)}$. Note that these vertices are pairwise nonadjacent, so $\alpha(G) \geqslant \frac{n}{\chi(G)}$. □

3 Graphs, Ramsey Theorem

Recall:

- $\alpha(G)$: Independent number = max ♯ pairwise nonadjacent vertices of G
- $\chi(G)$: Chromatic number = min ♯ colors s.t. $\exists$ a coloring of $V(G)$ is a proper coloring
- $n \leq \alpha(G)\chi(G)$

Definition. *Path $v_1 v_2 \cdots v_s$, where $v_i \sim v_{i+1}$ and $v_i \neq v_j, \forall i \neq j \in [s]$. If $v_1 = v_s$, call it a cycle. A graph G is* ***connected*** *if there is a path between any two vertices.*

Theorem 3.1. $|V(G)| = n$. *If for any* $x \in V(G)$, $\deg(x) \geq \frac{n-1}{2}$, *then* G *is connected.*

Proof. Take any different $x, y \in V(G)$. If $x \sim y$, then done.

If $x \nsim y$, since $\deg(x), \deg(y) \geq \frac{n-1}{2}$, there are at least $n-1$ edges joining x, y to $V(G) \setminus \{x, y\}$. Since $|V(G) \setminus \{x, y\}| = n-2$, by P-P, $\exists z \in V(G) \setminus \{x, y\}$, $z \sim x, z \sim y$. □

Remark:

(1) The condition above is best possible: e.g. n even, G is the union of two vertex disjoint complete graphs of $\frac{n}{2}$ vertices, each vertex has degree $\frac{n-2}{2}$, but G is disconnected.

(2) Define the **diameter** of G is the smallest number k, s.t. every two vertices are connected by a path with at most k edges. Then Theorem 1 says G has diameter at most two.

Fact(A party of six): Suppose a party has 6 participants. Participants may know each other or not. Then there must be 3 people such that any 2 know each other or any 2 don't know each other.

Proof. Construct a graph with vertices $[6]$, where $i \sim j$ iff i and j know each other. Then we need to show that there are 3 vertices in G which form a triangle or an independent set of size 3.

Consider vertex 1, by P-P, 1 is either adjacent to ≥ 3 vertices or nonadjacent to ≥ 3 vertices.

① Suppose 1 is adjacent to $2, 3, 4$. If one of the pairs $\{2, 3\}$, $\{2, 4\}$, $\{3, 4\}$ is adjacent, then we have a K_3. If not, $\{2, 3, 4\}$ is an independent set of size 3.

② Suppose 1 is nonadjacent to $2, 3, 4$. Similar arguments.

□

Definition. $\forall s, t \geq 1$, *let* $R(s, t)$ *denote the smallest integer* n, *s.t. in any graph with* n *or more vertices, there exists either a clique(a complete subgraph) with* s *vertices* K_s *or an independent set with* t *vertices* I_t.

Remark:

① $R(s, t) \leq L \iff$ any graph with L vertices has either a K_s or an I_t.

② $R(s, t) > M \iff \exists$ a graph with M vertices has neither K_s nor I_t.

Fact:

① $R(s, t) = R(t, s)$.

② $R(2, t) = R(t, 2) = t$.

③ $R(3,3) = 6$.

Theorem 3.2. *For $s \geq 2,\ t \geq 2$, $R(s,t) \leq R(s,t-1) + R(s-1,t)$.*

Proof. Let G be a graph on $n = R(s,t-1) + R(s-1,t)$ vertices. We need to prove any graph on n vertices has either a K_s or an I_t. Take an arbitrary vertex $x \in V(G)$. Let $S_x = \{y \in V(G) :\ x \sim y\}$ and $T_x = (V \setminus \{x\}) \setminus S_x$, then $|S_x| + |T_x| = n - 1 = R(s,t-1) + R(s-1,t) - 1$. By P-P, we have either $|T_x| \geq R(s,t-1)$ or $|S_x| \geq R(s-1,t)$.

① $|S_x| \geq R(s-1,t)$. Consider the induced subgraph $G[S]$: a graph on S, in which $v \sim w$ iff $v \sim w$ in G. Since $G[S_x]$ has at least $R(s-1,t)$ vertices, $G[S]$ has either a K_{s-1} or an I_t. Therefore $G[S_x \cup \{x\}]$ has either a K_s or an I_t.

② $|T_x| \geq R(s,t-1)$. Similar.

□

Theorem 3.3. $R(s,t) \leq \binom{s+t-2}{s-1} = \binom{s+t-2}{t-1}$.

Proof. By induction on $s+t$. $R(2,t) = t$, $R(s,2) = s$, true.

Assume the claim holds for $R(k,l)$ with $k + l < s + t$. Then $R(s,t) \leq R(s,t-1) + R(s-1,t) \leq \binom{s+t-3}{s-1} + \binom{s+t-3}{s-2} = \binom{s+t-2}{s-1}$.

□

Note: $2^{\frac{t}{2}} \leq R(t,t) \leq 2^{2t}$ $(Erdös 1947)$.

Theorem 3.4. *If $R(s,t-1), R(s-1,t)$ are even, then $R(s,t) \leq R(s,t-1) + R(s-1,t) - 1$*

Proof. $n = R(s,t-1) + R(s-1,t) - 1$, odd. We need to show any G on n vertices has a K_s or an I_t, $\forall x \in V$, let $S_x = \{y \in V(G) : x \sim y\}$ and $T_x = (V \setminus \{x\}) \setminus S_x$.

① If $\exists x$ s.t. $|S_x| \geq R(s-1,t)$ or $|T_x| \geq R(s,t-1)$, done

② $\forall x,\ |S_x| \leq R(s-1,t) - 1$ and $|T_x| \leq R(s,t-1) - 1$. $\because |S_x| + |T_x| = n - 1 = R(s-1,t) + R(s,t-1) - 2$, $\therefore |S_x| = R(s-1,t) - 1$, odd. Contradiction to the Handshaking Lemma.

□

Ex: Compute $R(3,4)$.

2-coloring version of Ramsey's theorem. Define a r-edge-coloring of K_n to be a coloring of edges of K_n by r colors. Then $R(s,t)$ denotes the smallest integer N s.t. any 2-edge-coloring of K_N has either a blue K_s or a red K_t.

Generalized Ramsey number $R_k(s_1, s_2, \ldots, s_k)$ is the smallest integer N such that any k-edge-coloring of K_N has a K_{s_i} in color i for some $i \in [k]$.

Ramsey Thm: $R_k(S_1, \cdots, S_k) \leq +\infty$.

4 Poset, chain, antichain, Pigeonhole Principle

Pigeonhole Principle and Poset

Definition. *We say y covers x if $x < y$ and $\nexists\ t$ such that $x < t < y$, denoted by $x \lhd y$.*

Fact. $\forall\ x, y \in P = (X, \leqslant)$, $x < y$ if and only if $\exists\ x_1, \cdots, x_k \in X$ where $k \geqslant 0$ s.t. $x \lhd x_1 \lhd \cdots \lhd x_k \lhd y$.

Proof. " $\Leftarrow$ " : trivial.
" $\Rightarrow$ " : $\forall\ x < y$, let $M_{xy} = \{t \in X : x < t < y\} = [x, y] \backslash \{x, y\}$.
Prove by induction on $|M_{xy}|$.
If $|M_{xy}| = 0$, then $x \lhd y$ holds.
Suppose the claim holds for any $x < y$ with $|M_{xy}| < n$. Consider $x < y$ with $|M_{xy}| = n \geqslant 1$.
Pick $t \in M_{xy}$. Consider M_{xt} and M_{ty}.
Since $M_{xt} \subsetneqq M_{xy}$ and $M_{ty} \subsetneqq M_{xy}$, we have $|M_{xt}| < n$ and $|M_{ty}| < n$.
By induction, $\exists\ x_1, \cdots, x_k \in X$ and $y_1, \cdots, y_l \in X$ such that $x \lhd x_1 \lhd \cdots \lhd x_k \lhd t$ and $t \lhd y_1 \lhd \cdots \lhd y_l \lhd y$.
So $x \lhd x_1 \lhd \cdots \lhd x_k \lhd t \lhd y_1 \lhd \cdots \lhd y_l \lhd y$. $\square$

Definition. *Hasse diagram of $P = (X, \leqslant)$ in the plane:*

- *$\forall\ x \in X$ is drawn as a node.*
- *$\forall\ x, y \in X$ with $x \lhd y$ is connected by a line segment.*
- *If $x \lhd y$, then x appears lower than y.*

Definition.

(1) $A \subset P = (X, \leqslant)$ is a chain if any two elements of A are comparable, i.e. A is totally ordered. Let $\omega(P)$ be the maximum size of a chain of P.

(2) $A \subset P = (X, \leqslant)$ is an antichain if any two elements of A are incomparable. Let $\alpha(P)$ be the maximum size of an antichain of P.

Remark. In Hasse diagram, $\omega(P)$ is called the height of P, and $\alpha(P)$ is called the width of P.

Fact. The set of minimal (or maximal) elements of $P = (X, \leqslant)$ forms an antichain.

Proof. $\forall$ finite poset $P = (X, \leqslant)$, we have $\alpha(P) \cdot \omega(P) \geqslant |X|$.
Let $P_1 = P$, $X_1 = X$, $M_1 =$\{minimal elements of P_1\}.
Let $X_2 = X_1 - M_1$, $P_2 = (X_2, \leqslant)$, $M_2 =$\{minimal elements of P_2\}.
Inductively, we can define a sequence of posets $P_i = (X_i, \leqslant)$, and $M_i =$\{minimal elements of P_i\}. $X_i = X - \bigcup\limits_{j=1}^{i-1} M_j$ for $i \leqslant l$, and $X_{l+1} = \emptyset$.

Each M_i is an antichain of P_i, thus it is also an antichain of P. So $|M_i| \leqslant \alpha(P)$.

Claim: $\forall\, x \in M_{i+1}$, $\exists\, y \in M_i$ such that $y < x$. (Since otherwise $x \in M_i$.)

So, we can find a chain $x_1 < x_2 < \cdots < x_l$ in P with $x_i \in M_i$, which implies $l \leqslant \omega(P)$.

Since $X = M_1 \sqcup M_2 \sqcup \cdots \sqcup M_l$, we have:

$$|X| = \sum_{i=1}^{l} |M_i| \leqslant \alpha(P) \cdot l \leqslant \alpha(P) \cdot \omega(P).$$

□

Theorem 4.1 (Dilworth)**.** *For any $P = (X, \leqslant)$ with $|X| \geqslant sr + 1$, there exists a chain of size $s + 1$ or an antichain of size $r + 1$.*

Proof. Suppose not, i.e. $\alpha(P) \leqslant r$ and $\omega(P) \leqslant s$.

Then by Theorem 1, $|X| \leqslant \alpha(P) \cdot \omega(P) \leqslant sr$, a contradiction. □

Definition. *Let $A = (a_1, \cdots, a_n)$ be a sequence of n different real numbers, and $B = (a_{i_1}, \cdots, a_{i_k})$ with $i_1 < \cdots < i_k$ be a subsequence. Call B an increasing subsequence (or decreasing subsequence) if $a_{i_1} \leqslant \cdots \leqslant a_{i_k}$ (or $a_{i_1} \geqslant \cdots \geqslant a_{i_k}$).*

Theorem 4.2 (Erdös–Szekeres)**.** *Let $A = (a_1, \cdots, a_n)$ be a sequence of n different real numbers. If $n \geqslant sr + 1$, then there exists an increasing subsequence of length $s + 1$ or a decreasing subsequence of length $t + 1$.*

Proof. Let $X = \{a_1, \cdots, a_n\}$. Define a poset $P = (X, \preccurlyeq)$, where $a_i \preccurlyeq a_j$ iff $a_i \leqslant a_j$ and $i \leqslant j$ (i.e. a_j appears after a_i in A).

Verify that P is indeed a poset:

(1) $a_i \preccurlyeq a_i$, $\forall\, i \in [n]$.

(2) If $a_i \preccurlyeq a_j$ and $a_j \preccurlyeq a_i$, then $a_i = a_j$.

(3) If $a_i \preccurlyeq a_j$ and $a_j \preccurlyeq a_l$, then $a_i \preccurlyeq a_l$.

In $P = (X, \preccurlyeq)$, a chain $a_{i_1} \prec \cdots \prec a_{i_k}$ is an increasing subsequence, since $a_{i_j} < a_{i_{j+1}}$ and $i_j < i_{j+1}$ for $j \in [k-1]$; an antichain $\{b_{i_1}, \cdots, b_{i_k}\}$ with $i_1 < \cdots < i_k$ is a decreasing subsequence, since b_{i_j} and $b_{i_{j+1}}$ are incomparable, which means $b_{i_j} < b_{i_{j+1}}$, $\forall\, j \in [k-1]$.

By Theorem 2, we can get the desired result. □

Rational approximation (Dirichlet, 1879)

Theorem 4.3. *Given any integer $n > 0$, $\forall\, x \in \mathbb{R}$, $\exists$ a rational number $\dfrac{p}{q}$ with $1 \leqslant q \leqslant n$ and $|x - \dfrac{p}{q}| < \dfrac{1}{nq} \leqslant \dfrac{1}{n}$.*

Proof. Let $\{x\} = x - \lfloor x \rfloor$ be the fractional part of x.

Consider $\{ax\}$ for $a = 1, \cdots, n+1$, which are $n+1$ real numbers in $[0, 1)$.

Put these numbers into n pigeonholes: $[0, \frac{1}{n})$, $[\frac{1}{n}, \frac{2}{n}), \cdots, [\frac{n-1}{n}, 1)$.

Then by P-P, there are 2 numbers, say $\{ax\}$ and $\{bx\}$ with $a > b$, are in the same interval. So $|\{ax\} - \{bx\}| < \frac{1}{n}$.

Since $ax - bx = (\lfloor ax \rfloor - \lfloor bx \rfloor) + \{ax\} - \{bx\}$, we have $\{ax\} - \{bx\} = (a-b)x - (\lfloor ax \rfloor - \lfloor bx \rfloor)$.

Let $q = a - b$, so $1 \leqslant q \leqslant n$, and let $p = \lfloor ax \rfloor - \lfloor bx \rfloor$.

So we have $|qx - p| < \frac{1}{n}$, i.e. $|x - p/q| < \frac{1}{nq}$. □

Subset without divisors

Question. How large a subset $S \subset [2n]$ can be such that for $\forall\, i, j \in S$, we have $i \nmid j$ and $j \nmid i$?

Example 4.4. $S = \{n+1, n+2, \cdots, 2n\}$ *with* $|S| = n$.

Theorem 4.5. *For any* $S \subset [2n]$ *with* $|S| \geqslant n+1$, $\exists\, i, j \in S$ *such that* $i \mid j$.

Proof. For each $k \in [n]$, let $S_k = \{x \in S : x$ can be written as $x = 2^i(2k-1)$ for some $i\}$. Then S can be partitioned into n sets: $S_1, S_2, \cdots, S_n$.
Since $|S| \geqslant n+1$, by P-P, $\exists\, k$ such that $|S_k| \geqslant 2$.

Assume $i, j \in S_k \subset S$, then either $i \mid j$ or $j \mid i$. □

5 Decomposition in chains/antichains, Sperner's Thm

Recall:

– Hassa diagram of a poset P $\longrightarrow$ Graph G

– $\alpha(P) =$ max size of antichain, $\alpha(P) = \alpha(G)$

– $w(P) =$ max size of chain, $w(P) = \chi(G)$

– $\alpha(P)w(P) \geq |X|$, $\alpha(G)\chi(G) \geq |X|$

Fact:

(1) Any finite poset P can be partitioned into $w(P)$ antichains.

(2) If a poset P has a chain(antichain) of size r, then it can not be partitioned into fewer than r antichains(chains).

Proof.

(1) Let Let $A_i = \{x \in P : \text{the longest chain with greatest element } x \text{ has size } i\}$. Then since $w(P) = r$, we have $A_i = \phi$ if $i \geq r+1$. So $P = A_1 \cup A_2 \cup \cdots \cup A_r$ is a partition (some of them may be empty). Show each A_i is an antichain. If not, say $x, y \in A_i, x < y$, then the largest chain with greatest element $x, x_1 < x_2 < \cdots < x_i = x$ can be prolonged to a longer chain $x_1 < x_2 < \cdots < x < y$, then the largest chain with greatest element y has size $\geq i+1$, i.e. $y \notin A_i$. Contradiction!

(2) Use pigeonhole principle.

□

Theorem 5.1 (Dilworth's Decomposition Thm)**.** *If $\alpha(P) = r$, then P can be partitions into r chains.*

Proof. Induction on $|P|$.

Assume the claim holds with any poset of size less than $|P|$.

Let a be a maximal element of P. Let $P' = P \setminus a$ and assume $\alpha(P') = s$, then P' can be partitioned into s chains $c_1, \cdots, c_s$

(1) $\alpha(P) = r = s+1$, then $c_1, \cdots, c_s, \{a\}$ form a partition of P

(2) $\alpha(P) = r = s$, let a_i be the maximal element in $\{x \in C_i : x \text{ belongs to some antichain of size } s\} \neq \phi$ and assume $a_i \in B_i$, where B_i is an antichain in P' of size s.
Then $A = \{a_1, \cdots a_s\}$ is an antichain of P'. (If not, $\exists a_i < a_j, i \neq j$, $\because \exists x \in B_j \cap c_i, \therefore x \leq a_i < a_j$, then $x \leq a_j \in B_j$, a contradiction.)
$\because \alpha(P) = s$, $\therefore a > a_i$ for some $i \in [s]$. Let $K = \{a\} \cup \{x \in c_i : \ x \leq a_i\}$, then K is a chain. $\alpha(P \setminus K) = s-1$ since K breaks all antichains of size of s. By assumption, $P \setminus K$ is a union of $s-1$ chains. Then P is a union of s chains.

□

Definition. *A set system (i.e. a family of subsets) $\mathscr{F}$ is an antichain with inclusion order (or sperner system) if $A \neq B \in \mathscr{F}$, neither $A \subset B$ nor $B \subset A$.*

Sperner's Thm. Let $\mathscr{F}$ be a set system over $[n]$, i.e. $\mathscr{F} \subset 2^{[n]}$. If $\mathscr{F}$ is an antichain, then $|\mathscr{F}| \leq \binom{n}{\lfloor \frac{n}{2} \rfloor}$.

Note.

(1) $P = (2^{[n]}, \subset)$, $\alpha(P) = \binom{n}{\lfloor \frac{n}{2} \rfloor}$.

(2) P can be partitioned into $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ symmetric chains.

LYM Inequality. $\mathscr{F}$ is an antichain over $[n]$, then $\sum_{A\in\mathscr{F}} \binom{n}{|A|}^{-1} \leq 1$.

Note. LYM Inequality $\Rightarrow$ Sperner's Thm: since $\binom{n}{|A|} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor}$.

Proof of LYM Inequality. A permutation$(x_1, x_2, \cdots, x_n)$, consider a maximal symmetric chain $\{\phi\}$ - $\{x_1\}$ - $\{x_1, x_2\}$ - $\cdots$ - $\{x_1, \cdots, x_n\}$.

Count $S = \{(A, C) : A \in \mathscr{F}, C$ is a a maximal symmetric chain, $A \in C\}$, fix A, $\exists\ |A|!(n-|A|)!$ permutations $(x_1, \cdots, x_n)$ s.t. $A = (x_1, \cdots, x_{|A|})$, then $\exists\ |A|!(n-|A|)!$ maximal symmetric chains. Since $\mathscr{F}$ is an antichain, then each maximal symmetric chain contains at most one $A \in \mathscr{F}$. Then

$$\sum_{A\in\mathscr{F}} |A|!(n-|A|)! = |S| \leq |\{\text{maximal symmetric chains}\}| = n!,$$

$$\Rightarrow \sum_{A\in\mathscr{F}} \binom{n}{|A|}^{-1} \leq 1.$$

□

Bollobás Thm: Let $A_1, \cdots, A_m$ and $B_1 \cdots, B_m$ be two sequence of sets,such that $A_i \cap B_j = \phi$ iff $i = j$. Then $\sum_{i=1}^{m} \binom{a_i + b_i}{a_i}^{-1} \leq 1$, where $|A_i| = a_i, |B_i| = b_i, i \in [m]$. In particular,if $a_i = a$ and $b_i = b$ for all $i \in [m]$, then $m \leq \binom{a+b}{a}$.

Note. Bollobás Thm $\Rightarrow$ LYM Inequality $\Rightarrow$ Sperner's Thm.

Proof. Let $\mathscr{F} = \{A_1, \cdots, A_n\} \subset 2^{[n]}$, $B_i = [n] \setminus A_i$, easily check $A_i \cap B_j = \phi$ iff $i = j$. By Bollobás Thm, $\sum_{i=1}^{m} \binom{n}{a_i}^{-1} \leq 1$.

□

Proof of Bollobás Thm. Let $X = (\cup_{i=1}^{m} A_i) \cup (\cup_{i=1}^{m} B_i)$, suppose $|X| = n$. $\forall A, B \subset X$, and $A \cap B = \phi$, say a permutation $(x_1, \cdots, x_n)$ of X separates the pair (A, B) if no element of B precedes an element of A, i.e. if $x_k \in A,\ x_l \in B$, then $k < l$.

We claim that each permutation separates at most one pair $(A_i, B_i), i \in [m]$. Indeed, suppose $(x_1, \cdots, x_n)$ separates $(A_i, B_i), (A_j, B_j), i \neq j$ and assume $\max\{k : x_k \in A_i\} \leq \max\{k : x_k \in A_j\}$. Since $(x_1, \cdots, x_n)$ separates (A_j, B_j), we have

$$\min\{k : x_k \in B_j\} > \max\{k : x_k \in A_j\} \geq max\{k : x_k \in A_i\},$$

$\Rightarrow A_i \cap B_j = \phi$, a contradiction.

Now count the number N_i of permutations separated one fixed pair say (A_i, B_i). First choose $a_i + b_i$ positions for $A \cup B$, there are $\binom{n}{a_i + b_i}$ choices. Then A occupy the first a_i positions, giving $a_i!$ choices for the order of A,

and $b_i!$ choices for the order of B, the remaining elements can be arranged in $(n-a_i-b_i)!$ ways. So $N_i = \binom{n}{a_i+b_i} a_i! b_i! (n-a_i-b_i)! = n! \binom{a_i+b_i}{a_i}^{-1}$.

Summing over all pairs, we have

$$\sum_{i=1}^{m} n! \binom{a_i+b_i}{a_i}^{-1} \leq n! \Rightarrow \sum_{i=1}^{m} \binom{a_i+b_i}{a_i}^{-1} \leq 1.$$

□

6 Intersecting Family, EKR Thm

Recall:

LYM Inequality: $\mathscr{F}$ is an antichain in $(2^{[n]}, \subseteq)$, then $\sum_{A \in \mathscr{F}} \binom{n}{|A|}^{-1} \leq 1$.

Proof. A maximal symmetric chain $\{\phi\}$ - $\{x_1\}$ - $\{x_1, x_2\}$ - $\cdots$ - $\{x_1, \cdots, x_n\}$, consider a permutation $\pi = (x_1, x_2, \cdots, x_n)$.

Double counting $S = \{(A, \pi) : A \in \mathscr{F}, \pi \in S_n,\ \pi \text{ contains } A\}$, where 'contain' means $A = \{x_1, \cdots, x_{|A|}\}$.

Fix A, $\sharp\pi$ contains $A = |A|!(n-|A|)!$

Fix π, since $\mathscr{F}$ is an antichain, π contains at most one $A \in \mathscr{F}$. Then

$$\sum_{A \in \mathscr{F}} |A|!(n-|A|)! = |S| = \sum_{\pi \in S_n} |\{A \in \mathscr{F} : \pi \text{ contains } A\}| \leq n!,$$

$$\Rightarrow \sum_{A \in \mathscr{F}} \binom{n}{|A|}^{-1} \leq 1.$$

□

Intersecting Family: A family of sets $\mathscr{F} \subset 2^{[n]}$ is intersecting if $\forall A, B \in \mathscr{F},\ |A \cap B| \geq 1$.

Example 6.1.

(1) $\mathscr{F} = \{A \in [n] :\ 1 \in a\},\ |\mathscr{F}| = 2^{n-1}$.

(2) for n odd, $\mathscr{F} = \{A \in [n] :\ |A| \geq \frac{n+1}{2}\},\ |\mathscr{F}| = \sum_{k=\frac{n+1}{2}}^{n} \binom{n}{k} = 2^{n-1}$.

Fact: For any intersecting family $\mathscr{F} \subset 2^{[n]} \implies |\mathscr{F}| \leq 2^{n-1}$, because $\mathscr{F}$ can not contain both A and $A^c = [n] \setminus A$.

Uniform case: $\mathscr{F} \subset \binom{[n]}{k}$, $\mathscr{F}$ is called $k-uniform$.

How large is $|\mathscr{F}|$ if $\mathscr{F}$ is intersecting?

Example 6.2.

(1) $\mathscr{F} = \{A \in \binom{[n]}{k} : \ 1 \in A\}, \ |\mathscr{F}| = \binom{n-1}{k-1}$*;*

(2) $k \geq \frac{n+1}{2}$, $\mathscr{F} = \binom{[n]}{k}, \ |\mathscr{F}| = \binom{n}{k}$.

Theorem 6.3 (Erdös-Ko-Rado Thm, *EKR*)**.** *If* $n \geq 2k, \mathcal{F} \subset \binom{[n]}{k}$ *is intersecting family, then* $|\mathcal{F}| \leq \binom{n-1}{k-1}$.

Proof. Doubling count (A, π), where $A \in \mathcal{F}$, π is a cyclic permutation vector and $A \in \mathcal{F}_\pi$ (which means that A appears as consecutive elements in π).

Fix A, $\sharp\pi = |A|!(n - |A|)! = k!(n-k)!$.

Fix π, $\forall A \in \mathcal{F} \cap \mathcal{F}_\pi, \exists 2k-2$ sets $B \in \mathcal{F}_\pi$ s.t. $B \cap A \neq \emptyset, \ B \neq A$. But they can be partitioned into k-1 pairs of disjoint subsets. Since $\mathcal{F}$ is intersecting, $\mathcal{F}$ contains at most one set of each pair.

$\Longrightarrow |\mathcal{F} \cap \mathcal{F}_\pi| \leq k$

$\Longrightarrow |\mathcal{F}| \cdot k!(n-k)! = \sum\limits_{A \in \mathcal{F}} k!(n-k)! = \sum\limits_{\pi} |\mathcal{F} \cap \mathcal{F}_\pi| \leq \sum\limits_{\pi} k = (n-1)!k$

$\Longrightarrow |\mathcal{F}| \leq \binom{n-1}{k-1}$

□

Theorem 6.4 (EKR Extremal Case)**.** *If* $n > 2k.$*, then the intersecting family* $\mathcal{F} \subset \dbinom{X}{k}$ *with* $|\mathcal{F}| = \dbinom{n-1}{k-1}$ *must be a star, i.e.* $|\mathcal{F}| = \{A \in \binom{[n]}{k} : \ i \in A\}$ *for some* $i \in [n]$.

Note: $n = 2k$, e.g. $\mathcal{F}$ contains exactly one set of (A, A^c), $|A| = k$, $|\mathcal{F}| = \dbinom{2k}{k}/2 = \binom{2k-1}{k-1}$.

Proof. See from the proof of EKR, if $|\mathcal{F}| = \dbinom{n-1}{k-1}$.

(1) $\forall$ cyclic π, $|\mathcal{F} \cap \mathcal{F}_\pi| = k$;

(2) if $\pi = (a_1, a_2, \cdots, a_n)$ cyclic vector, then $\mathcal{F} \cap \mathcal{F}_\pi = \{A_1, A_2, \cdots, A_k\}$, $A_j = \{a_j, \cdots, a_{j+k-1}\}$, $j = 1, \cdots, k$.

Fix π, let $\{a_k = A_1 \cap \cdots \cap A_k\}$. If all $A \in \mathcal{F}$, $a_k \in A$, then $\mathcal{F}$ is a star, done. If not, $\exists A_0 \in \mathcal{F}, a_k \notin A_0$, we will show that $\mathcal{F} = \dbinom{A_1 \cup A_k}{k}$, then $|\mathcal{F}| = \dbinom{2k-1}{k} = \dbinom{2k-1}{k-1} \leq \dbinom{n-1}{k-1}$ contradiction.

Claim1: $\forall B \in \binom{A_1 \cup A_k \setminus \{a_k\}}{k-1}$, $B \cup \{a_k\} \in \mathcal{F}$.

Proof: $B = B_1 \cup B_2$, where $B_1 \subset A_1$ and $B_2 \subset A_k$. Construct $\pi' = (A_1 \setminus (B_1 \cup \{a_k\}), B_1, a_k, B_2, A_k \setminus (B_2 \cup a_k))$. Since $A_1, A_k \in \mathcal{F}$, by 2, $B \cup \{a_k\} = B_1 \cup B_2 \cup \{a_k\} \in \mathcal{F}$

Claim2: $A_0 \subseteq (A_1 \cup A_k) \setminus \{a_k\}$.

Proof: If $A_0 \nsubseteq A_1 \cup A_k \setminus \{a_k\}, |A_0 \cap (A_1 \cup A_k \setminus \{a_k\})| \leq k-1$. $\exists B \subset (A_1 \cup A_k \setminus \{a_k\}) \setminus A_0, |B| = k-1$. By Claim1, $B \cup \{a_k\} \in \mathcal{F}$. But $A_0 \cap (B \cup \{a_k\}) = \emptyset$. Contradiction.

Claim3: $\binom{A_1 \cup A_k}{k} \subset \mathcal{F}$.

Proof: $\forall C \in \binom{A_1 \cup A_k}{k}$, $a_k \notin C$, $C \cap A_0 = \{b_1, b_2, \cdots, b_s\}$. $\pi = (\cdots b_1, \cdots, b_s \cdots)$, $C \in \mathcal{F}$.

Claim4: $\mathcal{F} \subset \binom{A_1 \cup A_k}{k}$.

Proof: If $\exists B \in \mathcal{F}.B \notin A_1 \cup A_k$, then $|B \cap (A_1 \cup A_k)| \leq k-1$. Then $\exists C \in \binom{A_1 \cup A_k}{k}, C \cap B = \emptyset$. Therefore, $C \notin \mathcal{F}$. But by Claim3, contradiction.

□

7 Hoffman's Theorem, Fisher's Inequality, design

Recall (EKR Theorem). If $n \geqslant 2k$, $\mathcal{F} \subset \binom{[n]}{k}$ is an intersecting family, then $|\mathcal{F}| \leqslant \binom{n-1}{k-1}$.

Definition. *Kneser graph $KG(n,k)$ for $n \geqslant 2k$ is a graph with vertex set $\binom{[n]}{k}$ such that $A, B \in \binom{[n]}{k}$, $A \sim B$ iff $A \cap B = \emptyset$.*

Fact.

(1) $\mathcal{F} \subset \binom{[n]}{k}$ is an intersecting family. $\iff$ $\mathcal{F}$ is an independent set in $KG(n,k)$.

(2) EKR Theorem $\iff \alpha(KG(n,k)) \leqslant \binom{n-1}{k-1}$

Definition. *$G = (V, E)$, $V = [n]$, the adjacency matrix of G $A_G = (a_{ij})_{n \times n}$ is defined by $a_{ij} = \begin{cases} 1, & \text{if } i \sim j, \\ 0, & \text{otherwise.} \end{cases}$*

Definition. *The real eigenvalues $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_n$ of A_G is called the eigenvalues of G. The eigenvectors $v_1, v_2, \cdots, v_n$ of A_G such that $\begin{cases} A_G v_i = \lambda_i v_i, \|v_i\| = 1 \text{ for all } i \in [n] \\ v_i \perp v_j \text{ for all } i \neq j. \end{cases}$ are called the orthogonal eigenvectors of G.*

Definition. *A graph G is called d-regular if each vertex has degree d.*

Theorem 7.1 (Hoffman's Theorem)**.** *If an n-vertex graph G is d-regular with eigenvalues $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_n$, then $\alpha(G) \leqslant n\frac{-\lambda_n}{\lambda_1-\lambda_n}$.*

Proof. Let $v_1, \cdots, v_n$ be the corresponding orthogonal eigenvectors of $\lambda_1, \cdots, \lambda_n$. Let I be any independent set of G. Let e_I be the column indicator vector of I, and write $e_I = \sum\limits_{i=1}^{n} \alpha_i v_i$, where $\alpha_i = \langle e_I, v_i \rangle$.

Since G is d-regular, we have $\lambda_1 = d$ (*prove this as an exercise!*) and $v_1 = (\frac{1}{\sqrt{n}}, \cdots, \frac{1}{\sqrt{n}})^T$. So $\alpha_1 = \langle e_I, v_1 \rangle = \frac{|I|}{\sqrt{n}}$.

Since I is an independent set of G, $e_I^T A_G e_I = \sum\limits_{i,j} x_i a_{ij} x_j = 0$, where we assume $e_I = (x_1, \cdots, x_n)^T$.

So we have

$$\begin{aligned}
0 = e_I^T A_G e_I = e_I^T \sum_{i=1}^{n} \alpha_i A_G v_i &= \sum_{i=1}^{n} \alpha_i \lambda_i \langle e_I, v_i \rangle \\
&= \sum_{i=1}^{n} \alpha_i^2 \lambda_i \geqslant \alpha_1^2 \lambda_1 + (\alpha_2^2 + \cdots + \alpha_n^2)\lambda_n \\
&= (\frac{|I|}{\sqrt{n}})^2 \lambda_1 + (|I| - (\frac{|I|}{\sqrt{n}})^2)\lambda_n
\end{aligned}$$

$$\Longrightarrow 0 \geqslant \frac{|I|^2}{n}\lambda_1 + (|I| - \frac{|I|^2}{n})\lambda_n = |I|(\frac{|I|}{n}\lambda_1 + \lambda_n - \frac{|I|}{n}\lambda_n)$$

$$\Longrightarrow \frac{|I|}{n}\lambda_1 + \lambda_n - \frac{|I|}{n}\lambda_n \leqslant 0 \qquad \Longrightarrow \frac{|I|}{n}(\lambda_1 - \lambda_n) \leqslant -\lambda_n$$

$$\Longrightarrow |I| \leqslant n\frac{-\lambda_n}{\lambda_1 - \lambda_n}$$

□

Lemma 7.2. *The eigenvalues of $KG(n,k)$ are $u_j = (-1)^j \binom{n-k-j}{k-j}$ of multiplicity $\binom{n}{j} - \binom{n}{j-1}$ for every $0 \leqslant j \leqslant k$.*

Proof. Omitted. (Refer to Chapter 9 of *GTM 207* Algebraic Graph Theory.) □

The second proof of EKR Theorem. Consider the eigenvalues of $KG(n,k)$, say $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_{\binom{n}{k}}$, where $\lambda_1 = \binom{n-k}{k} = u_0$ and $\lambda_{\binom{n}{k}} = -\binom{n-k-1}{k-1} = u_1$.

Then by Hoffman's bound, we have

$$\alpha(KG(n,k)) \leqslant \binom{n}{k}\frac{-\lambda_{\binom{n}{k}}}{\lambda_1-\lambda_{\binom{n}{k}}} = \binom{n}{k}\frac{\binom{n-k-1}{k-1}}{\binom{n-k}{k}+\binom{n-k-1}{k-1}} = \binom{n-1}{k-1}.$$

By Fact(2), we are done. □

Theorem 7.3 (Fisher's Inequality)**.** *Let $A_1, \cdots, A_m$ be distinct subsets of $[n]$ such that $|A_i \cap A_j| = k$ for some fixed $k \in [n]$, $\forall\ i \neq j$, then $m \leqslant n$.*

Proof. Consider the incidence matrix of $A_1, \cdots, A_m$, and let $v_1, \cdots, v_m$ be the m column vectors. Then it suffices to show that $v_1, \cdots, v_m$ are linearly independent in $\mathbb{R}$.

Assume, for the sake of contradiction, that $\exists\ \lambda_i \neq 0$ s.t. $\sum\limits_{i=1}^{m} \lambda_i v_i = 0$.

Since $\langle v_i, v_j\rangle = |A_i \cap A_j| = \begin{cases} |A_i| \geqslant k, & i = j, \\ k, & i \neq j, \end{cases}$ we have

$$\begin{aligned} 0 = \langle \sum_{i=1}^{m} \lambda_i v_i, \sum_{j=1}^{m} \lambda_j v_j\rangle &= \sum_{i=1}^{m} \lambda_i^2 \langle v_i, v_i\rangle + \sum_{1\leqslant i\neq j\leqslant m} \lambda_i\lambda_j\langle v_i, v_j\rangle \\ &= \sum_{i=1}^{m} \lambda_i^2 |A_i| + k \sum_{1\leqslant i\neq j\leqslant m} \lambda_i\lambda_j = \sum_{i=1}^{m} \lambda_i^2(|A_i|-k) + k(\sum_{i=1}^{m}\lambda_i)^2 \end{aligned}$$

Note that $|A_i| = k$ for at most one $i \in [m]$. (Otherwise $|A_i \cap A_j| < k$ for $|A_i|, |A_j| = k$ with $i \neq j$.) But there are at least two $\lambda_i \neq 0$. So $\sum\limits_{i=1}^{m} \lambda_i^2(|A_i|-k) > 0$, a contradiction. □

Designs

Definition. *A (v, k, λ) design over X is a pair (X, D), where*

- *X is a set of points. $|X| = v$.*
- *$D \subset \binom{X}{k}$ is called a set of blocks. $|D| = b$.*
- *$\forall$ pair of distinct points of X is contained in exactly λ blocks.*

Example 7.4. $X = [7],\ v = 7,\ k = 3,\ \lambda = 1,\ D = \{123, 147, 156, 345, 367, 257, 246\}$.

Fact. $b \geqslant v$.

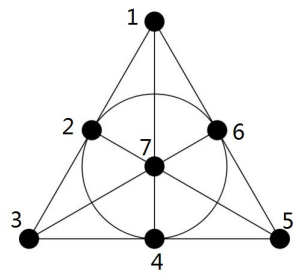


Figure 1: the Fano plane

Proof. $\forall\ x \in X$, define $A_x = \{B \in D :\ x \in B\}$.

Since each pair of distinct points of X is contained in exactly λ blocks, $|A_x \cap A_y| = \lambda$ for $x \neq y$.

By Fisher's Inequality, $v \leqslant b$. □

Definition. *If $b = v$, a (v, k, λ) design is called a symmetric design.*

Definition. *D is called r-regular if every point appears in exactly r blocks, and r is called replication number.*

Theorem 7.5. *(X, D) is a (v, k, λ) design with b blocks, then D is r-regular satisfying $r(k-1) = \lambda(v-1)$ and $bk = vr$.*

Proof. $\forall\ a \in X$, assume a occurs in r_a blocks.

Consider $S = \{(x, B) : B \in D;\ a, x \in B;\ x \neq a\}$.

- Since a is fixed, there are $v-1$ possibilities for x. For each chosen x, there are exactly λ blocks containing both x and a. Hence $|S| = (v-1)\lambda$.
- For each of the r_a blocks containing a, there are $k-1$ ways to choose $x \in B \setminus \{a\}$. Hence $|S| = r_a(k-1)$.

So, by double counting, we have $r_a(k-1) = (v-1)\lambda$ for $\forall\ a \in X$, which implies that r_a is independent of a, i.e. D is regular.

To prove $bk = vr$, consider $T = \{(x, B) : B \in D, x \in B\}$.

- $\forall\ x \in X$, we can choose B in r ways. Hence $|T| = vr$.
- $\forall\ B \in D$, we can choose x in k ways. Hence $|T| = bk$.

□

8 Difference Set, Finite Linear Space

Recall:

A (v, k, λ) design (X, D), $|X| = v$, $|D| = b$, r replication number satisfies each pair $\{x, y\} \subset X$ appears in exactly λ blocks.

- $b \geq v$, if $b = v$, called symmetric design
- $r(k-1) = \lambda(v-1)$ & $bk = rv$

Suppose G is an Abelian group of size v

Definition. $2 \leq k < v$, $\lambda \geq 1$, *A* (v, k, λ) ***difference set*** *(DS) is a k-subset* $D = \{d_1, d_2, \cdots, d_k\} \subseteq G$, *s.t. the collection of differences* $d_i - d_j$ $(i \neq j)$ *contains every element in* $G \setminus \{0\}$ *exactly* λ *times.*

Note. If $G = \mathbb{Z}_v$, call D is cyclic DS.

Example 8.1. $G = \mathbb{Z}_7$, *(7,3,1)-DS,* $D = \{0, 1, 3\}$

Fact:

① $\lambda(v-1) = k(k-1) \implies \lambda < k$.

② A **translate** of D is $a + D = \{a + d_1, a + d_2, \ldots, a + d_k\}$ for some $a \in G$. Then $a + D \neq D$ if $a \neq 0$.

Proof.

① Count # of differences in D.

② If $a + D = D$ for some $a \neq 0$, then $\exists$ a permutation π of $[k]$ satisfies that $\pi(i) \neq i$ and $d_i + a = d_{\pi(i)}$ for all $i \in [k]$. Then a is expressed as a difference $d_{\pi(i)} - d_i$ in k ways. But $k > \lambda$, contradiction.

□

Theorem 8.2. *If* D *is a* (v, k, λ) *difference set, then* $\{a + D : \ a \in G\}$ *are blocks of a symmetric* (v, k, λ) *design.*

Proof.

① $b = v = |G| \implies$ symmetric.

② $|i + D| = k$, $\forall i$.

③ Show any pair of points is contained in exactly λ blocks. $\forall x \neq y \in G$, assume $x - y = d \neq 0$, then $\{x, y\} \subset a + D \iff x = a + d_i, y = a + d_j$ for some $i \neq j \iff x - y = d_i - d_j = d \neq 0$. Since there are exactly λ pairs d_i, d_j s.t. $d_i - d_j = d$, and for each such pair, there are exactly one $a = x - d_i = y - d_j$ s.t. $\{x, y\} \subset a + D$.

□

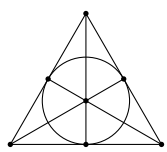
Theorem 8.3. *If* q *is a prime power,* $q \equiv 3\ (4)$, *then nonzero squares in* F_q *form a* $(q, \frac{q-1}{2}, \frac{q-3}{4})$ *DS.*

Proof. $G = F_q = \{0, \alpha^0, \alpha^1, \cdots, \alpha^{q-2}\},\ D = \{\alpha^0, \alpha^2, \cdots, \alpha^{q-3}\}$,then
$k = \dfrac{q-1}{2}$
$\because q \equiv 3\ (4),\ \therefore -1 = \alpha^{\frac{q-1}{2}} \notin D \Longrightarrow$ if $s \in D$, then $-s \notin D$
$\forall s \in D,\ \exists x, y \in D$ and $x - y = 1 \Longleftrightarrow \exists sx, sy \in D$ and $sx - sy = s \Longleftrightarrow \exists sx, sy \in D$ and $sy - sx = -s$. This means all nonzero space and nonsquares have the same # of representatives as a difference of two elements in D.
$\Longrightarrow \lambda = \dfrac{k(k-1)}{q-1} = \dfrac{\frac{q-1}{2}\frac{q-3}{2}}{q-1} = \dfrac{q-3}{4}$

□

Definition. *Finite linear space over a finite set X is a family α of subsets of X called lines such that:*

- *every line contains at least 2 points*
- *any 2 points are on exactly one line*

Example 8.4. 

Theorem 8.5. *If α is a finite linear space over X with $|\alpha| \geq 2$, then $|\alpha| \geq |X|$ with equality holds iff any two lines share exactly one point.*

Proof(Conway). Let $b = |\alpha| \geq 2,\ v = |X|$. $\forall x \in X$, let r_x be the replication number, i.e. # lines through x.

Choose a line $L \in \alpha$ and a point $x \notin L$. $\because \forall a \neq b \in L, \{x, a\} and \{x, b\}$ are on different lines, $\therefore r_x \geq |L|$.

Assume $b \leq v$, then $b(v - |L|) = bv - b|L| \geq bv - vr_x = v(b - r_x)$,
$\Longrightarrow$

$$\frac{1}{v - |L|} \leq \frac{b}{v(b - r_x)}$$

$$\begin{aligned} b = \sum_{L \in \alpha} 1 &= \sum_{L \in \alpha} \sum_{x \in X : x \notin L} \frac{1}{v - |L|} \leq \frac{b}{v} \sum_{L \in \alpha} \sum_{x \in X : x \notin L} \frac{1}{b - r_x} \\ &= \frac{b}{v} \sum_{x \in X} \sum_{L \in \alpha : x \notin L} \frac{1}{b - r_x} = \frac{b}{v} \sum_{x \in X} 1 = b \end{aligned}$$

This implies all inequalities are equalities so that $b = v$ and $r_x = |L|$ whenever $x \notin L$, i.e. any line containing x share one point with L.

$\therefore b \geq v$ with equality holds iff any two lines share exactly one point. □

9 Projective Plane, Resolvable Design, Affine Plane

Recall.

(1) A (v, k, λ) design $(X, \mathcal{D})$: $|X| = v$, $\mathcal{D} \subset \binom{X}{k}$ such that $\forall\ \{x, y\} \subset X$, $\{x, y\}$ appears in exactly λ blocks.
r: replication number.
$\Rightarrow r(k-1) = \lambda(v-1)$, $bk = vr$, $b = |\mathcal{D}| = \dfrac{\lambda v(v-1)}{k(k-1)} \geqslant v$.

(2) A (v, k, λ) difference set D.
$(G, \{a + D : a \in G\})$ is a (v, k, λ) design.

(3) Finite linear space $(X, \mathcal{L})$: $\mathcal{L} \subseteq 2^X$; $\forall\ L \in \mathcal{L}, |L| \geqslant 2$ such that $\forall\ \{x, y\} \subset X$ determine exactly one line.
$|\mathcal{L}| \geqslant 2 \Rightarrow |\mathcal{L}| \geqslant |X|$. Equality holds $\Longleftrightarrow$ Every two lines intersect in exactly one point.

Projective Plane ($PG(q)$)

Definition. *A projective plane of order $q \geqslant 1$ is a finite linear space with $q^2 + q + 1$ points, and each line has $q + 1$ points.*

Remark. *A projective plane of order q, denoted by $PG(q)$, is a $(q^2 + q + 1, q + 1, 1)$ symmetric design.*

Example 9.1. *(1) $q = 1$:*

(2) $q = 2$: Fano plane, $(7, 3, 1)$ design:

Proposition 9.2. *In $PG(q)$:*

(1) Any point lies on $q + 1$ lines.

(2) There are in total $q^2 + q + 1$ lines.

(3) Any two lines meet in a unique point.

Construction of $PG(q)$ for prime power $q \geqslant 2$.

Consider $\mathbb{F}_q^3$: 3-dim vector space over $\mathbb{F}_q$.
$V = \{(x_0, x_1, x_2) \in \mathbb{F}_q^3 \mid (x_0, x_1, x_2) \neq (0, 0, 0)\}$, then $|V| = q^3 - 1$.

(1) **points:** $[x_0, x_1, x_2] := \{(cx_0, cx_1, cx_2) : c \in \mathbb{F}_q \backslash \{0\}\}$. So there are $\dfrac{|V|}{q-1} = \dfrac{q^3-1}{q-1} = q^2+q+1$ points.

(2) **lines:** $L(a_0, a_1, a_2)$, where $(a_0, a_1, a_2) \in V$, is defined to be the set of points $[x_0, x_1, x_2]$ for which $a_0x_0 + a_1x_1 + a_2x_2 = 0$. There are q^2-1 solutions to this equation, so there are $\dfrac{q^2-1}{q-1} = q+1$ points in line $L(a_0, a_1, a_2)$.

(3) **CHECK any two points lie on a unique line:**
i.e. $\forall\ [x_0, x_1, x_2] \neq [y_0, y_1, y_2]$, $\exists!\ L(a_0, a_1, a_2)$ such that

$$\begin{cases} a_0x_0 + a_1x_1 + a_2x_2 = 0 \\ a_0y_0 + a_1y_1 + a_2y_2 = 0 \end{cases}.$$

Since $\begin{bmatrix} x_0 & x_1 & x_2 \\ y_0 & y_1 & y_2 \end{bmatrix}$ has rank 2, the solution space has dimension 1, i.e. $\exists!$ line $L(a_0, a_1, a_2)$ containing both $[x_0, x_1, x_2]$ and $[y_0, y_1, y_2]$.

Remark. $\forall\ q \geqslant 2$: *prime power,* $\exists\ (q^2+q+1, q+1, 1)$ *design:* $PG(q)$.

Conjecture (open). If q is not a prime power, $\exists\ PG(q)$?

Resolvable Design

Definition. *$(X, \mathcal{D})$ is a (v, k, λ) design, r : replication number. A parallel class is a set of blocks from $\mathcal{D}$ such that they partition X. ($\Rightarrow$ Each parallel class has $\dfrac{v}{k}$ blocks.)*
A partition of $\mathcal{D}$ into r parallel classes is called a resolution.
A design is said to be resolvable if it has a resolution.

Problem 1 (Kirkman's schoolgirl problem).
15 girls in a school walk out 3 abreast for 7 days in succession. Is it possible to arrange them daily so that no two shall walk twice abreast?
Solution. $\Leftrightarrow$ Find $(15, 3, 1)$ resolvable design.

Known results:

(1) $\exists\ (v, 3, 1)$ design for $v \equiv 1, 3 \pmod 6$: Steiner Triple System (STS)

(2) $\exists\ (v, 3, 1)$ resolvable design for $v \equiv 3 \pmod 6$: Kirkman Triple System (KTS)

Example 9.3. *KTS (9,3,1)*

123	147	159	168
456	258	267	249
789	369	348	357

Problem 2. A football league of $2n$ teams. Is it possible to arrange a schedule such that they play in $2n-1$ days, and on each day every team plays one match?
Solution. The answer is a resolvable $(2n, 2, 1)$ design $(X, \mathcal{D})$.
Let $X = \{*\} \cup [2n-1]$, $\mathcal{D} = \binom{X}{2}$, $\mathcal{D} = \mathcal{D}_1 \cup \cdots \cup \mathcal{D}_{2n-1}$ where $\mathcal{D}_i := \{\{i, *\}\} \cup \{\{a, b\} : \ a + b \equiv 2i \pmod{2n-1}\}$ for $i \in [2n-1]$.

(1) $\forall\ a \neq b \in X$: If $a = *$, then $\{a, b\} \in \mathcal{D}_b$. If $a \neq *$, $b \neq *$, then $\exists!\ i \in [2n-1]$ s.t. $a + b \equiv 2i \pmod{2n-1}$, i.e. $\exists!\ \mathcal{D}_i$ s.t. $\{a, b\} \in \mathcal{D}_i$. $\Longrightarrow$ $\mathcal{D} = \mathcal{D}_1 \cup \cdots \cup \mathcal{D}_{2n-1}$ is a partition.

(2) Given $i \in [2n-1]$, CHECK $\mathcal{D}_i$ is a parallel class. $\forall\ a \in X$: If $a = *$, then the unique block in D_i containing a is $\{i, *\}$. If $a \neq *$, then $\exists!\ b$ s.t. $a + b \equiv 2i \pmod{2n-1}$, i.e. the unique block in D_i containing a is $\{a, b\}$.

Affine Plane $AG(q)$

Definition. *Affine plane $AG(q)$ is a finite linear space such that all lines form a $(q^2, q, 1)$ design.*

Remark. *For $AG(q)$, we have: $r = \dfrac{q^2-1}{q-1} = q+1$, $b = \dfrac{vr}{k} = q^2 + q$.*

Construction of $AG(q)$ from $PG(q)$.
Construction 1.
$(X, \mathcal{L})$: $PG(q)$. Then $|X| = q^2 + q + 1$, $|\mathcal{L}| = q^2 + q + 1$, $|L| = q + 1$.
Fix $L_0 \in \mathcal{L}$. Let $X' = X \setminus \{L_0\}$, $\mathcal{L}' = \{L \setminus L_0 : L \in \mathcal{L}, L \neq L_0\}$. Then it's easy to see that $(X', \mathcal{L}')$ is $AG(q)$.
Next, we show resolvability. $\forall\ x \in L_0$, let $\mathcal{L}_x = \{L \setminus \{x\} : L_0 \neq L \in \mathcal{L}, x \in L\}$. Then $\mathcal{L}_x$ is a parallel class. Since in $PG(q)$, every $L \neq L_0$ intersects L_0 in exactly one point, we know that $\bigcup_{x \in L_0} \mathcal{L}_x$ is a partition of $\mathcal{L}'$.
Construction 2.
$q \geqslant 2$: prime power. Consider $\mathbb{F}_q$.
Let $X = \mathbb{F}_q \times \mathbb{F}_q$. Let $\mathcal{L}$ be the set of all lines of the form

$$L(a, b) = \{(x, y) \in X : y = ax + b\},\ a, b \in \mathbb{F}_q,$$

and

$$L(c) = \{(c, y) \in X : y \in \mathbb{F}_q\},\ c \in \mathbb{F}_q.$$

Then $(X, \mathcal{L})$ is $AG(q)$.

Proof. $|X| = q^2$, $|L| = q$. For $\forall\ (x_1, y_1) \neq (x_2, y_2) \in X$,

(1) If $x_1 = x_2$, then the unique line containing them is $L(x_1)$.

(2) If $x_1 \neq x_2$, then

$$\begin{cases} y_1 = ax_1 + b \\ y_2 = ax_2 + b \end{cases}$$

has a unique solution (a, b), i.e. the unique line containing them is $L(a, b)$. Next, show resolvability. $\{L(a, b) : b \in \mathbb{F}_q\}$ $(a \in \mathbb{F}_q)$, $\{L(c) : c \in \mathbb{F}_q\}$ is a resolution. (Note that $r = q + 1$.) $\square$

Fact. Any $AG(q)$ is resolvable.

10 Linear Algebra method, Frankl-Wilson Thm

Recall:

(1) intersecting family $\mathcal{F} \subset 2^{[n]}$: $\forall A, B \in \mathcal{F}, |A \cap B| \neq 0$

(2) Fisher inequality: let $A_1, \cdots, A_m$ be subsets of $[n]$, $|A_i \cap A_j| = k$, for some fixed $k \in [n]$, then $m \leq n$

Definition. *$\mathcal{F} \subset 2^{[n]}$, $L \subset \{0, 1, \cdots\}$ be a finite set of integers, say $\mathcal{F}$ is $L-$intersecting if $|A \cap B| \in L$ for $\forall A \neq B \in \mathcal{F}$*

Theorem 10.1 (Frankl-Wilson)**.** *If $\mathcal{F}$ is an $L-$intersecting family over $[n]$, then $|\mathcal{F}| \leq \sum_{i=0}^{|L|} \binom{n}{i}$.*

Note: The bound is best possible: Let $L = \{0, 1, \cdots, k\}$ $\mathcal{F} = \{\emptyset\} \cup \binom{[n]}{1} \cup \cdots \binom{[n]}{k+1}$

Note:

(1) Let F is a field $(\mathbb{C}, \mathbb{R}, \mathbb{Q}, \mathbb{F}_q)$, Ω is a set. $F^\Omega = \{\text{function:}\Omega \to F\}$ is a linear space over F. A set of functions $f_1, \cdots, f_m$ is linearly independent if $\forall$ combination $\lambda_1 f_1 + \cdots + \lambda_m f_m = 0, \lambda_i \in F$, then $\lambda_i = 0,\ i \in [m]$

(2) Consider $\{f(x_1, \cdots, x_n)$ polynomials with degree$\leq d\}$, then each of f is combination of $x_1^{t_1} \cdots x_n^{t_n}$, with $t_1 + \cdots + t_n \leq d, t_i \geq 0$. The dimension is $\sum_{i=0}^{d} \binom{i+n-1}{i} = \binom{n+d}{d}$

Lemma 10.2 (Independence criterion)**.** *If $i \in [m]$, let $f_i : \Omega \to F$ (where F is a field) be functions and $v_i \in \Omega$ such that*

(1) $f_i(v_i) \neq 0, \forall i \in [m]$.

(2) $f_i(v_j) = 0, \forall 1 \leq j < i \leq m$.

Then $f_1, \cdots, f_m$ are linearly independent in the function space F^Ω.

Proof. Assume $\exists$ not all zeros $\lambda_i,\ i \in [m]$, s.t.

$$\lambda_1 f_1 + \lambda_2 f_2 + \cdots + \lambda_m f_m = 0$$

Suppose j is the smallest index such that $\lambda_j \neq 0$. then

$$0 = \lambda_{j+1} f_{j+1}(v_j) + \lambda_{j+2} f_{j+2}(v_j) + \cdots + \lambda_n f_n(v_j) = -\lambda_i f_j(v_j) \neq 0$$

□

Proof. Suppose $\mathcal{F} = \{A_1, \cdots, A_m\}$,with $|A_1| \leq \cdots \leq |A_m|$, and $L = \{l_1, \cdots l_s\}$. Let v_i be the indicator vector of $A_i,\ i \in [m]$, then $< v_i, v_j >= |A_i \cap A_j| = l_k$ for some $k \in [s]$. For $i = 1, \cdots, m$, define f_i with n variables $\vec{x} = (x_1, \cdots, x_n)$ by

$$f_i : \mathbb{R}^n \to \mathbb{R}.$$

$$f_i(\vec{x}) = \prod_{k: l_k < |A_i|} (< v_i, \vec{x} > - l_k).$$

$f_i(v_i) \neq 0, \forall i \in [m]$
If $1 \leq j < i \leq m,\ < v_i, v_j >= |A_i \cap A_j| < |A_i| \implies f_i(v_j) = 0$
By **Lemma 10.2**, $f_1, \cdots, f_m$ are linearly independent over $\mathbb{R}$.
$\because f_i$ are polynomials of degree at most s, $\therefore m \leq \sum_{i=0}^{s} \binom{i+n-1}{i}$, but we can do it better!
Define new polynomials $\bar{f}_i$ form f_i by replacing all term $x_i^{r_i}$ $(r_i \geq 1)$ by x_i. Since v_i are $0-1$ vectors, we have $\bar{f}_i(v_j) = f_i(v_j), \forall i, j$, so $\bar{f}_i, \cdots, \bar{f}_m$ are linearly independent, who lie in a space with basis $x_1^{r_1} \cdots x_n^{r_n}$ with $r_1 + \cdots + r_n \leq s$ and $r_i \in \{0, 1\} \implies m \leq \sum_{i=0}^{s} \binom{n}{i}$. □

Theorem 10.3. *Let p be a prime and $L \subset \mathbb{Z}_p = \{0, 1, \cdots, p-1\}$.Assume $\mathcal{F} = \{A_1, \cdots A_m\} \subset 2^{[n]}$ such that*

(a) $|A_i| \notin L \pmod p, \forall i \in [m]$.

(b) $|A_i \cap A_j| \in L \pmod p, \forall i \neq j$.

Then $|\mathcal{F}| \leq \sum_{i=0}^{|L|} \binom{n}{i}$

Hint: $f_i : \mathbb{R}^n \to \mathbb{F}_p,\ f_i(x) = \prod_{l \in L} (< v_i, x > - l) \pmod p, i \in [m]$.

Note: Consider $p = 2, L = \{0\}$, then $|A_i|$ is odd and $|A_i \cap A_j|$ is even($\forall i \neq j$), $|\mathcal{F}| \leq n + 1$ (Actually $|\mathcal{F}| \leq n$, which will be proved in **Odd/Even Town**)

Ramsey number $R(s, t)$ = least integer N s.t. any graph on N vertices has either a K_s or an I_t

(1) $2^{\frac{t}{2}} < R(t, t) < \binom{2t-2}{t-1} < 2^{2t}$

(2) $R(t,t) > (t-1)^2$. (Homework 11.8)

(3) $R(s,t) > \Omega(t^3)$ 1972,Zsigmond Nagy

(4) $R(t,t) > t^{\Omega(\ln t/\ln\ln t)}$. 1977,Frankl, 1981 F&Wlison.

Theorem 10.4. *For any prime p, there is a graph G on $n = \binom{p^3}{p^2-1}$ vertices s.t. the size of maximum clique or maximum independent set is $\leq \sum_{i=0}^{p-1}\binom{p^3}{i}$.*

Proof. Let $G = (V,E)$ be as follows

- $V = \binom{[p^3]}{p^2-1}$
- for $A, B \in V, A \sim B$ iff $|A \cap B| \neq p-1 \pmod p$.

(i) consider the cliques $A_1, \cdots, A_m \in V, |A_i| = p^2-1 = p-1 \pmod p$, $|A \cap B| \neq p-1 \pmod p$ means $|A \cap B| \in L \pmod p$, where $L = \{0, 1, \cdots p-2\} \subset \mathbb{Z}_p$.
By **Thm 10.3**, we have $m \leq \sum_{i=0}^{p-1}\binom{p^3}{i}$.

(ii) consider the independent sets $B_1, \cdots, B_s$, $|B_i \cap B_j| = p-1 \pmod p$, so $|B_i \cap B_j| \in \{p-1, 2p-1, \cdots, p(p-1)-1\} = L^* \subset \mathbb{Z}_{\geq 0}$. and $|L^*| = p-1$,
By **Thm10.1**, we have $s \leq \sum_{i=0}^{p-1}\binom{p^3}{i}$.

□

Corollary 10.5. $R(t+1, t+1) \geq t^{\Omega(\ln t/\ln\ln t)}$

proof(sketch) :.

Let $t = \sum_{i=0}^{p-1}\binom{p^3}{i}$, $n = \binom{p^3}{p^2-1} \Longrightarrow R(t+1,t+1) > n$
Recall $(\frac{n}{k})^k \leq \binom{n}{k} \leq (\frac{en}{k})^k \Longrightarrow$
$t \sim \binom{p^3}{p} \sim (\frac{p^3}{p})^p = p^{2p}$
$n \sim \binom{p^3}{p^2} \sim p^{p^2}$
$\ln t \sim p\ln p$, $\ln\ln t \sim \ln(p\ln p) = \ln p + \ln\ln p \sim \ln p$
$p \sim \frac{\ln t}{\ln\ln t}$, $n \sim (p^{2p})^{\frac{p}{2}} = t^{\Omega(\ln t/\ln\ln t)}$ □

11 Odd/Even-town, Linear Algebra Method (about polynomials)

Recall.

(1) (Frankl-Wilson Theorem) $\mathcal{F} \subseteq 2^{[n]}$ is an L-intersecting family, then $|\mathcal{F}| \leqslant \sum_{i=0}^{|L|}\binom{n}{i}$.

(2) Let p be a prime and $L \subset \mathbb{Z}_p = \{0, 1, \cdots, p-1\}$. Assume $\mathcal{F} = \{A_1, \cdots A_m\} \subseteq 2^{[n]}$ such that

(a) $|A_i| \pmod p \notin L, \forall\, i \in [m]$;

(b) $|A_i \cap A_j| \pmod p \in L, \forall\, i \neq j$.

Then $|\mathcal{F}| \leqslant \sum\limits_{i=0}^{|L|} \binom{n}{i}$

(3) (Application) $\exists$ graph on $\binom{p^3}{p^2-1}$ vertices such that the size of maximum clique or maximum independent set is $\leqslant \sum\limits_{i=0}^{p-1} \binom{p^3}{i}$.

$\Longrightarrow R(t+1, t+1) \geqslant t^{\Omega(\ln t/\ln\ln t)}$.

Odd/Even-town. Let $\mathcal{F} \subseteq 2^{[n]}$ *s.t.* $|A|$ is odd for all $A \in \mathcal{F}$, and $|A \cap B|$ is even for $\forall\, A \neq B \in \mathcal{F}$. Then $|\mathcal{F}| \leqslant n$.

Proof. $\forall\, A \in \mathcal{F}$, let $\mathbf{e_A}$ be the indictor vector of A. View $\mathbf{e_A} \in \mathbb{F}_2^n$, then

$$\begin{aligned} &< \mathbf{e_A}, \mathbf{e_A} >= 1, \quad \forall\, A \in \mathcal{F} \\ &< \mathbf{e_A}, \mathbf{e_B} >= 0, \quad \forall\, A \neq B \in \mathcal{F} \end{aligned}$$

Assume $\exists\, \alpha_A \in \mathbb{F}_2 = \{0, 1\}$ *s.t.* $\sum\limits_{A \in \mathcal{F}} \alpha_A \mathbf{e_A} = \mathbf{0}$, then

$$0 =< \sum_{A \in \mathcal{F}} \alpha_A \mathbf{e_A}, \mathbf{e_B} >= \alpha_B < \mathbf{e_B}, \mathbf{e_B} >= \alpha_B, \quad \forall\, B \in \mathcal{F}$$

So $\mathbf{e_A},\ A \in \mathcal{F}$ are linearly independent, which implies $|\mathcal{F}| \leqslant n$. □

Corollary 11.1. $R(t+1, t+1) > \binom{t}{3} \sim t^3$.

Proof. Define $G = (V, E)$ as follows:

$$V = \binom{[t]}{3}, \quad A \sim B \text{ iff } |A \cap B| = 1.$$

Consider a clique $A_1, \cdots, A_m$, then $|A_i \cap A_j| = 1, \forall\, i \neq j$. By Fisher's Inequality, $m \leqslant t$.
Condsider an independent set $B_1, \cdots B_s$, then $|B_i| = 3$ (*Odd*), $|B_i \cap B_j| = 0$ *or* 2 (*Even*). By Odd/Even-town, $s \leqslant t$. □

Even/Odd-town. Let $\mathcal{F} \subseteq 2^{[n]}$ *s.t.* $|A|$ is even for all $A \in \mathcal{F}$, and $|A \cap B|$ is odd for $\forall\, A \neq B \in \mathcal{F}$. Then $|\mathcal{F}| \leqslant n$.

Proof. For $\forall\ A \in \mathcal{F}$, define $A' = A \cup \{n+1\}$. Define $\mathcal{F}' = \{A' : A \in \mathcal{F}\} \subseteq 2^{[n+1]}$. Then, by Odd/Even-town, $|\mathcal{F}| = |\mathcal{F}'| \leqslant n+1$.
Assume $|\mathcal{F}| = n+1$. For $\forall\ A \in \mathcal{F}$, let $\mathbf{e_A} \in \mathbb{F}_2^n$ be the indictor vector of A. Then $\mathbf{e_A}$, $A \in \mathcal{F}$ are linearly dependent, *i.e.* $\exists$ not all zero $\alpha_A \in \mathbb{F}_2 = \{0,1\}$ *s.t.* $\sum\limits_{A\in\mathcal{F}} \alpha_A \mathbf{e_A} = \mathbf{0}$.
Note that

$$< \mathbf{e_A}, \mathbf{e_A} >= 0, \quad \forall\ A \in \mathcal{F}$$
$$< \mathbf{e_A}, \mathbf{e_B} >= 1, \quad \forall\ A \neq B \in \mathcal{F}$$

So

$$0 =< \sum_{A\in\mathcal{F}} \alpha_A \mathbf{e_A}, \mathbf{e_B} >= \sum_{A\in\mathcal{F}: A\neq B} \alpha_A = \sum_{A\in\mathcal{F}} \alpha_A - \alpha_B, \quad \forall\ B \in \mathcal{F}$$

$\Longrightarrow \alpha_B = \sum\limits_{A\in\mathcal{F}} \alpha_A = 1$, $\forall\ B \in \mathcal{F}$ (Since α_B are not all zero.)
$\Longrightarrow 1 = n+1$ in $\mathbb{F}_2$. $\quad\Longrightarrow n$ is even.
Consider $\mathcal{F}^c = \{A^c = [n]\backslash A : A \in \mathcal{F}\}$, then

- $|A^c|$ is even, $\forall\ A \in \mathcal{F}$
- $|A^c \cap B^c| = n - |A \cup B| = n - (|A| + |B| - |A \cap B|)$ is odd, $\forall\ A \neq B \in \mathcal{F}$

So $\mathcal{F}^c$ is also Even/Odd-town & $|\mathcal{F}^c| = |\mathcal{F}| = n+1$.
Repeat the previous proof, we have

$$\sum_{A\in\mathcal{F}} \mathbf{e_{A^c}} = \mathbf{0}.$$

Combine with $\sum\limits_{A\in\mathcal{F}} \mathbf{e_A} = \mathbf{0}$,
$\Longrightarrow \mathbf{0} = \sum\limits_{A\in\mathcal{F}} (\mathbf{e_A} + \mathbf{e_{A^c}}) = \sum\limits_{A\in\mathcal{F}} \mathbf{1} = (n+1)\mathbf{1} = \mathbf{1}$, a contradiction.
Therefore, $|\mathcal{F}| \leqslant n$. $\square$

F : field. $F[x_1, \cdots x_n]$ ={polynomial $f : F^n \to F$}.

Definition. *Say polynomial f vanishes on $E \subset F^n$ if $f(x_1, \cdots, x_n) = 0, \forall\ (x_1, \cdots, x_n) \in E$.*

f is zero polynomial if and only if all coefficients are 0.

Note. If $n = 1$, univariate $f(x) \neq 0$, $deg(f) \leqslant d$, then f has at most d roots.

(1) $f \neq 0$ vanishes on S, then $|S| \leqslant deg(f)$.

(2) If $\nexists\ f \neq 0$ that vanishes on S, then $|S| > deg(f)$.

Lemma 11.2. *Given $E \subset F^n$, $|E| < \binom{n+d}{d}$, then $\exists\ 0 \neq f \in F[x_1, \cdots, x_n]$ with $deg(f) \leqslant d$ that vanishes on E.*

Proof. Let $V_d = \{f \in F[x_1, \cdots, x_n] : deg(f) \leqslant d\}$, then $dim\ (V_d) = \binom{n+d}{d}$.
Let $F^E = \{$all functions from E to $F\} = \{$all vectors over F of length $|E|\}$. *i.e.* $u \in F^E$, $u = (u_a)_{a \in E}$, $u_a \in F$.
Define a map: $V_d \to F^E,\ f \mapsto (f(a))_{a \in E}$.
Note that F^E is a linear space, $dim\ (F^E) = |E| < \binom{n+d}{d} = dim\ (V_d)$. So $\exists\ f_1 \neq f_2 \in V_d$ such that $(f_1(a))_{a \in E} = (f_2(a))_{a \in E}$.
$\Longrightarrow ((f_1 - f_2)(a))_{a \in E} = 0$, *i.e.* $0 \neq f_1 - f_2 \in V_d$ vanishes on E. □

Lemma 11.3. *$\forall\ 0 \neq f \in \mathbb{F}_q[x_1, \cdots, x_n]$ with $deg(f) = d < q$ has at most dq^{n-1} roots.*

Proof. $n = 1$: univariate case, OK.
Assume $n \geqslant 2$. Write $f = g + h$, where g is homogenous of degree d and $deg(h) \leqslant d - 1$. Then $g \neq 0$, $\exists\ \omega \in \mathbb{F}_q^n \backslash \{0\}$ such that $g(\omega) \neq 0$.
$\forall\ u \in \mathbb{F}_q^n$, let $L_u = \{u + t\omega : t \in \mathbb{F}_q\}$ (line). $\Longrightarrow\ |L_u| = q$.
If $v \notin L_u$, then $L_u \cap L_v = \emptyset$. (Since otherwise, $u + t\omega = v + t'\omega \Longrightarrow v = u + (t - t')\omega \in L_u$.) Hence $\mathbb{F}_q^n$ is partitioned into $q^n/q = q^{n-1}$ disjoint lines.
Now, it remains to show that f has at most d roots in each line.
Let $P_u(t) = f(u + t\omega)$, then $deg(P_u) \leqslant d$ and a root of P_u corresponds to a root of f in line L_u.

$$[t^d]P_u(t) = [t^d]f(u + t\omega) = [t^d]g(u + t\omega) = [t^d]g(t\omega) = g(\omega) \neq 0$$

$\Longrightarrow P_u \neq 0$.
$\Longrightarrow P_u$ has at most d roots.
$\Longrightarrow f$ has at most d roots in line L_u. □

Lemma 11.4. *$\forall\ S \subset F$, $|S| \geqslant d$, $\forall\ 0 \neq f \in F[x_1, \cdots, x_n]$ of degree d. Then f has at most $d|S|^{n-1}$ roots in $S^n = S \times \cdots \times S$.*

Proof. Prove by induction on n.
$n = 1$: univariate case, OK.
Assume $n \geqslant 2$. Write $f = f_0 + f_1 x_n + \cdots + f_t x_n^t$, $t \leqslant d$, $f_t \neq 0$, $f_i \in F[x_1, \cdots, x_{n-1}]$, $deg(f_i) \leqslant d - i$.
Now estimate the number of points $(a, b) \in S^{n-1} \times S$ *s.t.* $f(a, b) = 0$.

(1) $f_t(a) = 0$. Since $deg(f_t) \leqslant d - t$, by assumption, f_t has at most $(d-t)|S|^{n-2}$ roots in S^{n-1}. So there are at most $(d-t)|S|^{n-2} \cdot |S| = (d-t)|S|^{n-1}$ points $(a, b) \in S^{n-1} \times S$ *s.t.* $f(a, b) = 0$ and $f_t(a) = 0$.

(2) $f_t(a) \neq 0$. Given such $a \in S^{n-1}$, let $g_a(x_n) = f(a, x_n)$, then $deg(g_a) \leqslant t$. So $g_a \neq 0$ has at most t roots in S. *i.e.* Given $a \in S^{n-1}$ satisfying $f_t(a) \neq 0$, $\exists$ at most t elements $b \in S$ *s.t.* $f(a, b) = g_a(b) = 0$. Since there are at most

$|S|^{n-1}$ such a, the number of points $(a,b) \in S^{n-1} \times S$ *s.t.* $f(a,b) = 0$ and $f_t(a) \neq 0$ is at most $t|S|^{n-1}$.

Together, there are at most $d|S|^{n-1}$ points $(a,b) \in S^{n-1} \times S$ such that $f(a,b) = 0$. □

12 Combinatorial Nullstellensatz

Recall:

- $0 \neq f \in \mathbb{F}_q[x_1, \cdots, x_n]$, deg$f$=$d$, then f has $\leq dq^{n-1}$ roots in $\mathbb{F}_q^n$
- $0 \neq f \in F[x_1, \cdots, x_n]$, deg$f$=$d$, then f has $\leq d|S|^{n-1}$ roots in S^n

Lemma 12.1. *Let $f \in F[x_1, \cdots, x_n]$ be a polynomial,and let t_i be the maximum degree of x_i in f. Let $S_i \subset F$ with $|S_i| \geq t_i + 1$. If $f(x) = 0, \forall x \in S_1 \times \cdots \times S_n$, then $f \equiv 0$.*

Proof. Induction on n. $n = 1$ is true.
Assume the claim holds for $n-1$ variables. Write $f = f_0 + f_1 x_n^1 + f_2 x_n^2 + \cdots + f_{t_n} x_n^{t_n}$, where $f_i \in F[x_1, \cdots, x_{n-1}]$. In each f_i, the maximum degree of x_j is $\leq t_j,\ j \in [n-1],\ i \in \{0, 1, \cdots, t_n\}$.
For any given $a \in S_1 \times S_2 \times \cdots S_{n-1}$. Let $g_a(x_n) = f(a, x_n)$, $\deg(g_a) = t_n$.
$\forall x_n \in S_n, g_a(x_n) = f(a, x_n) = 0$ and $|S_n| \geq t_n + 1 \Longrightarrow g_a \equiv 0$
$g_a(x_n) = f_0(a) + f_1(a)x_n^1 + \cdots + f_{t_n}(a)x_n^{t_n} \Longrightarrow f_0(a) = f_1(a) = \cdots = f_{t_n}(a) = 0, \forall a \in S_1 \times S_2 \times \cdots S_{n-1}$.
By assumption, $f_i = 0, i \in [n-1] \Longrightarrow f = 0$ □

Theorem 12.2. *(Nullstellensatz) Let $f \in F[x_1, \cdots, x_n]$, and let $S_1, \cdots, S_n$ be nonempty subsets of F and $f(x) = 0, \forall x \in S_1 \times \cdots \times S_n$, then exist polynomials $h_1, \cdots h_n \in F[x_1, \cdots x_n]$ such that $deg(h_i) \leq deg(f) - |S_i|$ and $f = \sum_{i=1}^{n} h_i \prod_{s \in S_i} (x_i - s)$*

Proof. Let $t_i = |S_i| - 1$, and $g_i(x_i) = \prod_{s \in S_i} (x_i - s) = x_i^{t_i+1} - \sum_{j=0}^{t_i} a_{ij} x_i^j$.

In f, replace x_1^d $(d \geq t_1 + 1)$ by $x_1^{d-(t_1+1)}(g_1(x_1) + \sum_{j=0}^{t_1} a_{1j} x_1^j)$, we get $f = h_1(x_1, \cdots, x_n) g_1(x_1) + f_1$, where max deg of x_1 in f_1 is $\leq t_1$ and degh_1 $\leq$deg$f$-$|S_1|$. Repeat this step for $x_i^{t_i+1},\ i \in [2, n]$, we get $f = \sum_{i=1}^{n} h_i g_i + \bar{f}$. In $\bar{f}$, each x_j has deg $\leq t_j$ and $\bar{f} = f = 0, \forall x \in S_1 \times \cdots S_n \Longrightarrow \bar{f} \equiv 0$. □

Theorem 12.3. *(Combinatorial Nullstellensatz) Let $f \in F[x_1, \cdots, x_n]$ be a polynomial of degree d. Suppose $[x_1^{t_1} x_2^{t_2} \cdots x_n^{t_n}] f \neq 0$ and $\sum_{i=1}^{n} t_i = d$. If $S_i \subset F$ with $|S_i| \geq t_i + 1, i \in [n]$, then $\exists x \in S_1 \times \cdots S_n$ for which $f(x) \neq 0$.*

Proof. By contradiction. Assume $|S_i| = t_i + 1, i \in [n]$ and $f(x) = 0$ for all $x \in S_1 \times \cdots S_n$. By **Thm12.2**, write $f = \sum_{i=1}^n h_i g_i$, where $\deg h_i \leq d - |S_i|$ and $g_i(x_i) = \prod_{s \in S_i}(x_i - s)$, then $f(x) = \sum_{i=1}^n h_i(x) x_i^{t_i+1} +$(terms of degree $< d$). By assumption, $[x_1^{t_1} x_2^{t_2} \cdots x_n^{t_n}]f$ on LHS is nonzero, but it is impossible to have such a monomial on RHS. □

Application of Combinatorial Nullstellensatz

Theorem 12.4. *(Chevalley-Warning) Let p be a prime and $f_1, \cdots, f_m \in \mathbb{F}_p[x_1, \cdots, x_n]$. If $\sum_{i=1}^m deg f_i < n$, and $f_1, \cdots, f_m$ have a common root $(c_1, \cdots, c_n)$, then they have another common root.*

Proof. By contradiction. Assume $(c_1, \cdots, c_n)$ is the only common root. Define $f(x_1, \cdots, x_n) = \prod_{i=1}^m (1 - f_i(x_1, \cdots, x_n)^{p-1}) - \delta \prod_{j=1}^n \prod_{c \in F_p, c \neq c_j} (x_j - c)$, where δ is chosen s.t. $f(c_1, \cdots, c_n) = 0$. i.e. $\delta = \frac{1}{\prod_{j=1}^n \prod_{c \in F_p, c \neq c_j}(c_j - c)}$
$\forall (s_1, \cdots, s_n) \in \mathbb{F}_p^n$ and $(s_1, \cdots, s_n) \neq (c_1, \cdots, c_n)$,
$\exists i \in [m]$ s.t. $f_i(s_1, \cdots, s_n) \neq 0$ in $\mathbb{F}_p$. By Fermat's little Theorem, $f_i(s_1, \cdots, s_n)^{p-1} \equiv 1 (mod p) \implies$ the first product on RHS is zero. It is easy to check that the second term is also zero, so $f_i(x_1, \cdots, x_n) = 0$, $\forall (x_1, \cdots, x_n) \in \mathbb{F}_p^n$.
Now check the degree of f. In the first product,the degree$\leq \sum_{i=1}^m deg f_i \cdot (p-1) < n(p-1) \implies \deg f = n(p-1)$, and the monomial $x_1^{p-1} x_2^{p-1} \cdots x_n^{p-1}$ has coefficient $-\delta \neq 0$. Let $S_i = \mathbb{F}_p$, then apply Combinatorial Nullstellensatz, $\exists x \in F_p^n, s.t. f(x) \neq 0$, a contradiction. □

Recall: $A = (a_{ij})_{n \times n}$, $per(A) = \sum_{(i_1, \cdots, i_n)} a_{1,i_1} a_{2,i_2} \cdots a_{n,i_n}$, where $(i_1, \cdots, i_n)$ runs over all permutations of [n].

Theorem 12.5. *(Permanent Lemma) Let $b \in F^n$ and $S_1, \cdots, S_n$ be subsets of F, and $|S_i| \geq 2$. If the $per(A) \neq 0$, then there $\exists x \in S_1 \times \cdots S_n$ such that Ax differs from b in all coordinates.*

Proof. Define $f = \prod_{i=1}^n (\sum_{i=1}^n a_{ij} x_j - b_i)$, need to show $\exists x, s.t. f(x) \neq 0$.
$\deg f = n$ and $[x_1 \cdots x_n]f = per(A) \neq 0$. Since $|S_i| \geq 2, i \in [n]$, then apply Combinatorial Nullstellensatz. □

Corollary 12.6. *If $per(A) \neq 0$, then for any $b \in F^n$, there is a subset of columns of A whose sum differs from b in all coordinates.*

Hint: Let $S_i = \{0, 1\}$

Theorem 12.7. *Let $G = (V, E)$ be a graph, no loops but multiple edges allowed. p is a prime, if average degree $> 2p - 2$, max degree $\leq 2p - 1$, then G contains a p-regular subgraph.*

Proof. To be continued. □

13 Combinatorial Nullstellensatz, Sumset, Zero-Sum Set

Recall (Combinatorial Nullstellensatz). Let $f \in F[x_1, \cdots, x_n]$ be a polynomial of degree d. Suppose $[x_1^{t_1} x_2^{t_2} \cdots x_n^{t_n}]f \neq 0$ and $\sum\limits_{i=1}^{n} t_i = d$. If $S_i \subset F$ with $|S_i| \geqslant t_i + 1$, $i \in [n]$, then $\exists\ x \in S_1 \times \cdots \times S_n$ *s.t.* $f(x) \neq 0$.

Theorem 13.1. *Let $G = (V, E)$. G has no loops but multiple edges are allowed. Let p be a prime. If average degree of $G > 2p - 2$, max degree of $G \leqslant 2p - 1$, then G contains a p-regular subgraph.*

Proof. Associate $\forall\ e \in E$ with a variable x_e.
Define $f = \prod\limits_{v \in V} [1 - (\sum\limits_{e \in E} a_{v,e} x_e)^{p-1}] - \prod\limits_{e \in E} (1 - x_e) \in \mathbb{F}_p[x_1, \cdots, x_{|E|}]$, where $a_{v,e} = 1$ if $v \in e$ and $a_{v,e} = 0$ if $v \notin e$.
Note that degree in the first product is $(p-1)|V|$, and degree in the second product is $|E|$. By Handshaking Lemma, average degree $\dfrac{\sum_{v \in V} d(v)}{|V|} = \dfrac{2|E|}{|V|} > 2p - 2 \Longrightarrow (p-1)|V| < |E|$. So $deg(f) = |E|$, and $[\prod\limits_{e \in E} x_e]f = (-1)^{|E|+1} \neq 0$.
Apply Combinatorial Nullstellensatz with $S_i = \{0, 1\}$, $t_e = 1$ for $\forall\ e \in E \Longrightarrow \exists\ x = (x_e)_{e \in E} \in \{0, 1\}^{|E|}$ *s.t.* $f(x) \neq 0$.
Now consider the subgraph H consisting of all edges $e \in E$ with $x_e = 1$. If $x = (0, \cdots, 0)$, $f(x) = 0$. So $x \neq (0, \cdots, 0)$. $\Longrightarrow H$ is not empty.
So

$$0 \neq f(x) = \prod_{v \in V} [1 - (\sum_{e \in E} a_{v,e} x_e)^{p-1}],$$

which implies $(\sum\limits_{e \in E} a_{v,e} x_e)^{p-1} \neq 1$ for $\forall\ v \in V$.
By Fermat's Little Theorem, $\sum\limits_{e \in E} a_{v,e} x_e \equiv 0 \pmod p$ for $\forall\ v \in V$.
$\Longrightarrow \sum\limits_{e \in E : v \in e} x_e = \sum\limits_{e \in E} a_{v,e} x_e \equiv 0 \pmod p$ for $\forall\ v \in V$
$\Longrightarrow$ Each vertex has degree $0 \pmod p$ in H.
Since the maximum degree $\leqslant 2p - 1$, all positive degrees are precisely p, *i.e.* H is p-regular. $\square$

<u>**Additive Combinatorics**</u>

Definition. *(sumset)*$A + B = \{a + b : a \in A, b \in B\}$.

Theorem 13.2. *(Cauchy-Davenport)If p is a prime, and A, B are two non-empty subsets of $\mathbb{F}_p$, then $|A + B| \geqslant min\ \{p, |A| + |B| - 1\}$.*

Proof. (1) If $|A| + |B| - 1 \geqslant p$, then $|A| + |B| \geqslant p + 1$, which implies $|A \cap B| \neq \emptyset$.
For $\forall\ x \in \mathbb{F}_p$, $|\{x\} - B| = |B|$. So $A \cap (\{x\} - B) \neq \emptyset$.
$\Longrightarrow \exists\ a \in A$, $a \in \{x\} - B$.
$\Longrightarrow \exists\ b \in B$, $a = x - b$, *i.e.* $x = a + b$.
$\Longrightarrow A + B = \mathbb{F}_p$. So $|A + B| \geqslant p$.

(2) If $|A|+|B|-1 \leqslant p-1$, then $|A|+|B| \leqslant p$. We need to show $|A+B| \geqslant |A|+|B|-1$.
Assume $|A+B| \leqslant |A|+|B|-2 \leqslant p-2$, then $\exists\, C \subset \mathbb{F}_p$ with $|C| = |A|+|B|-2$ *s.t.* $A+B \subset C$.
Define $f = \prod_{c \in C} (x_1 + x_2 - c) \in \mathbb{F}_p[x_1, x_2]$, then $f(x_1, x_2) = 0$ for $\forall\ (x_1, x_2) \in A \times B$ and $deg(f) = |C| = |A|+|B|-2$.
Let $t_1 = |A|-1$, $t_2 = |B|-1$, then $[x_1^{t_1} x_2^{t_2}]f = \binom{|C|}{|A|-1} = \binom{|A|+|B|-2}{|A|-1} \not\equiv 0 \pmod{p}$.
By Combinatorial Nullstellensatz, $\exists\ (x_1, x_2) \in A \times B$ *s.t.* $f(x_1, x_2) \neq 0$, a contradiction.

□

Zero-Sum Set

Question 1. Any sequence $a_1, \cdots, a_n$ of n integers contains a non-empty consecutive subsequence $a_i, a_{i+1}, \cdots, a_{i+m}$ whose sum is $0 \pmod{n}$.

Proof. Assume there are n holes labeled from 0 to $n-1$.
Consider n sequences: $(a_1), (a_1, a_2), \cdots, (a_1, a_2, \cdots, a_n)$. If the sum of a sequence is $i \pmod{n}$, put it into the i-th hole.
If the 0-th hole is not empty, we're done.
Suppose not. Then by Pigeonhole Principle, there are two sequences in the same hole, say, $(a_1, \cdots, a_{i-1})$ and $(a_1, \cdots, a_{i-1}, a_i, \cdots, a_{i+m})$, which means they have the same sum $\pmod{n}$. Then the subsequence $(a_i, \cdots, a_{i+m})$ has sum $0 \pmod{n}$. □

Question 2. Given $n > 0$, what is the smallest N such that any sequence of N integers contains a subsequence of n integers (not necessarily consecutive) whose sum is $0 \pmod{n}$?

Example. Consider $(a_1, \cdots, a_{2n-2})$ where $a_i = 0$ for $i = 1, \cdots, n-1$ and $a_i = 1$ for $i = n, \cdots, 2n-2$. $\Longrightarrow N > 2n-2$.

Theorem 13.3. *p is a prime, then any integer sequence of length $2p-1$ contains a subsequence of length p whose sum is $0 \pmod{p}$.*

<u>Proof 1:</u> Assume $a_1 \leqslant a_2 \leqslant \cdots \leqslant a_{2p-1}$.
If $\exists\ i \in [p-1]$ such that $a_i = \cdots = a_{i+p-1}$, we're done.
Suppose not. Define $A_i = \{a_i, a_{i+p-1}\}$ for $i \in [p-1]$. Then by Cauchy-Davenport Theorem, we have $|A_1 + A_2 + \cdots + A_{p-1}| \geqslant \min\{p, |A_2 + \cdots + A_{p-1}| + 1\} \geqslant \min\{p, |A_3 + \cdots + A_{p-1}| + 2\} \geqslant \cdots \geqslant \min\{p, |A_{p-1}| + p - 2\} = p$. Hence $A_1 + A_2 + \cdots + A_{p-1} = \mathbb{F}_p$. So $\exists\ a_{i_j} \in A_j$ such that $-a_{2p-1} = a_{i_1} + a_{i_2} + \cdots + a_{i_{p-1}}$. □

<u>Proof 2:</u> Define $f_1 = \sum_{i=1}^{2p-1} a_i x_i^{p-1}$ and $f_2 = \sum_{i=1}^{2p-1} x_i^{p-1} \in \mathbb{F}_p[x_1, \cdots, x_{2p-1}]$, then $f_1(0) = f_2(0) = 0$, and $deg(f_1) + deg(f_2) = 2(p-1) < 2p-1$. So $\exists\ x = (x_1, \cdots, x_{2p-1}) \neq 0$ such that $f_1(x) = f_2(x) = 0$. Here, we use the fact that if $f_1, \cdots, f_m \in F[x_1, \cdots, x_n]$, $\sum_{i=1}^{m} deg(f_i) < n$, and $(c_1, \cdots, c_n)$ is a common root of all f_i, then $\exists$ another common root.
Let $I = \{i \in [2p-1] : x_i \neq 0\}$. Then $f_2(x) = 0 \Longrightarrow |I| \equiv 0 \pmod p \Longrightarrow |I| = p$, and $f_1(x) = 0 \Longrightarrow \sum_{i \in I} a_i \equiv 0 \pmod p$. □

Theorem 13.4. *(generalization of theorem 13.3)n is an integer, then any integer sequence of length $2n-1$ contains a subsequence of length n whose sum is $0 \pmod n$.*

Proof. Prove by induction on the number of primes in n.
If $n = p$, we're done. (*i.e.* Theorem 13.3)
Assume $n = pm \geqslant 2p$, and Theorem 13.4 holds for m.
Consider sequence $a_1, \cdots, a_{2p-1}$, and apply Theorem 13.3 for p, we get $I_1 \subset \{a_1, \cdots, a_{2p-1}\}$ with $|I_1| = p$ and $\sum_{i \in I_1} a_i \equiv 0 \pmod p$.
Now consider sequence $\{a_1, \cdots, a_{2p-1}\} \backslash I_1, a_{2p}, \cdots, a_{2n-1}$, and apply Theorem 13.3 for p again, we get I_2 with $|I_2| = p$, $I_2 \cap I_1 = \emptyset$ and $\sum_{i \in I_2} a_i \equiv 0 \pmod p$.
Repeat this process until we can't do it any more, we get disjoint $I_1, \cdots, I_l$ with $|I_j| = p$ and $\sum_{i \in I_j} a_i \equiv 0 \pmod p$ for $\forall\ j \in [l]$.
Claim: $l \geqslant 2m-1$.
Proof of Claim: If $l \leqslant 2m-2$, then $(2n-1) - lp \geqslant (2pm-1) - (2m-2)p = 2p-1$, which means we can get I_{l+1}, a contradiciton.
Let $b_j = \dfrac{\sum_{i \in I_j} a_i}{p}$ for $j \in [l]$, then b_j is an integer.
Since $l \geqslant 2m-1$, by assumption, $\exists\ J \subset [2m-1]$ with $|J| = m$ such that $\sum_{j \in J} b_j \equiv 0 \pmod m$.
$\Longrightarrow \sum_{j \in J} \sum_{i \in I_j} \dfrac{a_i}{p} = \sum_{j \in J} b_j \equiv 0 \pmod m$
$\Longrightarrow \sum_{j \in J} \sum_{i \in I_j} a_i \equiv 0 \pmod{pm}$ *i.e.* $\sum_{j \in J} \sum_{i \in I_j} a_i \equiv 0 \pmod n$ □

14 Probabilistic Methods, 2-colorable

<u>Probabilistic Methods</u>

- **Ideas:**

① Image we need to find some combinatorial object satisfying certain property, call them "good" objects. We consider a random object. If the probability that the random object is "good" is positive (i.e. Pr[the object is good]>0), then we say there must exist "good" objects.

② To compute Pr[good object]>0, we often prove Pr[bad object]<1.

Definition. *A finite probability space is* $(\Omega, \Pr)$*, where* Ω *is a finite set and* $\Pr: \Omega \to [0,1]$ *such that* $\sum_{x\in\Omega} \Pr[x] = 1$.

- $A \subseteq \Omega$ *is called an event,* $\Pr[A] = \sum_{x\in A} \Pr[x]$.
- $\Pr[\emptyset] = 0$, $\Pr[\Omega] = 1$.
- $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B] \le \Pr[A] + \Pr[B]$*(union bound).* *"=" holds iff* $\Pr[A \cap B] = 0$.

Theorem 14.1. *Let* n, s *be integers satisfying* $\binom{n}{s}2^{1-\binom{s}{2}} < 1$*, then* $R(s,s) > n$.

Proof. We need to find a 2-edge-coloring of K_n such that it has NO monochromatic clique K_s. Consider a random 2-edge-coloring of K_n: each edge is colored blue or red, each with probability $\frac{1}{2}$, independent of other edges.

Let A be the event that the defined K_n has a monochromatic K_s (need to show $\Pr[A] < 1$).
For each $B \in \binom{[n]}{s}$, let A_B be the event that K_n has a monochromatic K_s with vertex set B. Then $A = \bigcup_{B \in \binom{[n]}{s}} A_B$.

By union bound, $\Pr[A] = \Pr[\bigcup_{B \in \binom{[n]}{s}} A_B] \le \sum_{B \in \binom{[n]}{s}} \Pr[A_B] = \binom{n}{s}2^{1-\binom{s}{2}} < 1$.
Thus $\Pr[\bar{A}] = 1 - \Pr[A] > 0$, i.e. the probability that K_n has NO monochromatic K_s is positive. So there must exist a 2-edge-coloring of K_n such that it has NO monochromatic K_s.
$\Longrightarrow R(s,s) > n$ □

Corollary 14.2. $R(k,k) \ge \frac{1}{\mathrm{e}\sqrt{2}} k 2^{\frac{k}{2}}$.

Proof. Let $n = \frac{1}{\mathrm{e}\sqrt{2}} k 2^{\frac{k}{2}} \left(\frac{\mathrm{e}}{2}\right)^{\frac{1}{k}}$. Recall $\binom{n}{k} < \frac{n^k}{k!}$ and $k! > \mathrm{e}\left(\frac{k}{\mathrm{e}}\right)^k$. We have

$$\binom{n}{k}2^{1-\binom{k}{2}} < \frac{n^k}{\mathrm{e}\left(\frac{k}{\mathrm{e}}\right)^k}2^{1-\binom{k}{2}} = \left(\frac{\mathrm{e}n}{k}\right)^k \left(\frac{2}{\mathrm{e}}\right) 2^{-\binom{k}{2}}.$$

Since $n = \frac{1}{\mathrm{e}\sqrt{2}} k 2^{\frac{k}{2}} \left(\frac{\mathrm{e}}{2}\right)^{\frac{1}{k}}$, we have $\binom{n}{k}2^{1-\binom{k}{2}} < 1$. Then $R(k,k) > n \ge \frac{1}{\mathrm{e}\sqrt{2}} k 2^{\frac{k}{2}}$. □

Definition. *A* ***tournament*** *of* n *vertices is an orientation of* K_n*. Say vertex* x *beats vertex* y *if* $x \to y$ *or* $(x,y) \in E$*.* ***Property*** $\mathbf{P_k}$ *:* $\forall$ *k-subset* $A \subset V$*, there is a vertex who beats all vertices in* A*.*

Question: For $\forall k \ge 2$, does there exist a tournament T with property P_k?

Theorem 14.3. *If $n \geq k^2 2^{k+1}$, then $\exists$ a tournament T of n players with property P_k.*

Proof. Consider a random tournament of K_n over n. For any $i < j$, the arc $i \to j$ occurs with probability $\frac{1}{2}$ independently. Let B be the event that T doesn't have P_k. For $A \in \binom{[n]}{k}$, let B_A be the event that $\forall x \in [n] \setminus A$, x can not beat every vertices in A. Then $B = \bigcup_{A \in \binom{[n]}{k}} B_A$.

Let $B_{A,x}$ be the event that x can not beat all vertices in A. Then $B_A = \bigcap_{x \in [n] \setminus A} B_{A,x}$. Clearly, $\Pr[B_{A,x}] = 1 - \left(\frac{1}{2}\right)^k$.

Note that arcs between x and A are disjoint, so all events $B_{A,x}$ are independent for $x \in [n] \setminus A$. $\Longrightarrow \Pr[B_A] = \prod_{x \in [n] \setminus A} \left(1 - \frac{1}{2^k}\right) = \left(1 - \frac{1}{2^k}\right)^{n-k}$.

By union bound, $\Pr[B] \leq \sum_{A \in \binom{[n]}{k}} \Pr[B_A] = \binom{n}{k} \left(1 - \frac{1}{2^k}\right)^{n-k}$.

Check $\binom{n}{k} \left(1 - \frac{1}{2^k}\right)^{n-k} < 1$ by following steps:

(i) $n > 2^k \cdot k + 1$, $\binom{n}{k} \left(1 - \frac{1}{2^k}\right)^{n-k}$ is decreasing.

(ii) $\binom{n}{k} \left(1 - \frac{1}{2^k}\right)^{n-k} < \frac{n^k}{k!} e^{-(\frac{1}{2})^k (n-k)} < n^k e^{-\frac{n}{2^k}}$

(iii) $n = k^2 2^{k+1}$, $n^k e^{-\frac{n}{2^k}} < 1$

□

Definition. *$\mathcal{F} \subseteq 2^{[n]}$, say $\mathcal{F}$ is **2-colorable** if $\exists$ a function $f: [n] \to \{$blue, red$\}$, s.t. every set $A \in \mathcal{F}$ is not monochromatic.*

Note: $\mathcal{F} \subseteq \binom{[n]}{2}$, $\mathcal{F}$ is 2-colorable iff $G = ([n], \mathcal{F})$ is bipartite.

Theorem 14.4. *Every k-uniform family $\mathcal{F}$ with $|\mathcal{F}| < 2^{k-1}$ is 2-colorable.*

Proof. Let $\mathcal{F}$ be any fixed k-uniform family over finite set $X = [n]$. Consider a random function $f: X \to \{$blue, red$\}$, s.t.$\forall x \in X$ is colored by blue or red with probability $\frac{1}{2}$, and the coloring of different elements are independent. Let B be the event that f is "bad", i.e. $\exists\ A \in \mathcal{F}$ is monochromatic.

For $A \in \mathcal{F}$, let B_A be the event that A is monochromatic. So $B = \bigcup_{A \in \mathcal{F}} B_A$. $\forall A \in \mathcal{F}$, $\Pr[B_A] = 2^{1-k}$. Hence $\Pr[B] \leq \sum_{A \in \mathcal{F}} \Pr[B_A] = |\mathcal{F}| 2^{1-k} < 1$. So $\Pr[\bar{B}] > 0$, i.e. $\exists$ a 2-coloring s.t. no $A \in \mathcal{F}$ is monochromatic. □

Note:

(i) $\mathcal{F} = \binom{[2k]}{k}$ is not 2-colorable

(ii) Let $B(k)$ be the minimum possible number of sets in a k-uniform family which is not 2-colorable, then $2^{k-1} \leq B(k) \leq \binom{2k}{k} < 2^{2k}$. But 2^{2k} is not good enough and we can prove the following Thm.

Theorem 14.5. *If k is sufficiently large, then there exists a k-uniform family $\mathcal{F}$ such that $|\mathcal{F}| \leq k^2 2^k$ and $\mathcal{F}$ is not 2-colorable.*

Proof. To be continued. □

15 Probabilistic method, 2-colorable, Linearity of Expectation, sum-free set, dominating set

Recall.

- $\mathcal{F} \subset \binom{[n]}{k}$ is 2-colorable if $\exists\ f : [n] \to \{\text{blue, red}\}$ *s.t.* NO $A \in \mathcal{F}$ is monochromatic.
- $B(k) = \min |\mathcal{F}|$ *s.t.* $\mathcal{F}$ is not 2-colorable
- $|\mathcal{F}| < 2^{k-1}$ is 2-colorable, and

$$2^{k-1} \leqslant B(k) \leqslant \binom{2k}{k}.$$

Theorem 15.1. *If k is sufficiently large, then there exists a k-uniform $\mathcal{F}$ s.t. $|\mathcal{F}| \leqslant k^2 \cdot 2^k$ and $\mathcal{F}$ is not 2-colorable.*

Proof. Let $X = [r]$. Choose b k-subsets from $\binom{[r]}{k}$ independently at random. Specifically, pick a k-subset A_1 from $\binom{[r]}{k}$ with probability $\frac{1}{\binom{r}{k}}$, and repeat this process b times to get $\mathcal{F} = \{A_1, \cdots, A_b\}$ (allow repetition).
Let B be the event that $\mathcal{F}$ is 2-colorable. It suffices to prove $\Pr[B] < 1$.
Let B_χ be the event that $\mathcal{F}$ is 2-colorable under coloring χ, then $B = \cup_{\chi \in 2^{[r]}} B_\chi$.

$$\begin{aligned}
\Pr[B_\chi] &= \Pr[\ \bigcap_{i=1}^{b} \{A_i \text{ is not monochromatic under } \chi\}\ ] \\
&= \prod_{i=1}^{b} \Pr[A_i \text{ is not monochromatic under } \chi] \\
&= \prod_{i=1}^{b} (1 - \Pr[A_i \text{ is monochromatic under } \chi])
\end{aligned}$$

Suppose χ has s red points and $r - s$ blue points, then

$$\begin{aligned}
\Pr[A_i \text{ is monochromatic under } \chi] &= \frac{\binom{s}{k} + \binom{r-s}{k}}{\binom{r}{k}} \\
&\overset{\text{Jensen's Inequality}}{\geqslant} \frac{2\binom{r/2}{k}}{\binom{r}{k}} \triangleq p
\end{aligned}$$

So, $\Pr[B_\chi] \leqslant (1-p)^b$ and

$$Pr[B] \leqslant \sum_{\chi \in 2^{[r]}} (1-p)^b = 2^r(1-p)^b \leqslant 2^r \cdot e^{-pb} = e^{r\ln 2 - pb}$$

Note that $p = \frac{1}{2^{k-1}} \prod_{i=0}^{k-1} (1 - \frac{i}{r}\frac{1}{1-\frac{i}{r}})$ and $1 - \frac{i}{r}\frac{1}{1-\frac{i}{r}} \sim 1 - \frac{i}{r} + O(\frac{i^2}{r^2})$, so $p \leqslant \frac{1}{2^{k-1}} \prod_{i=0}^{k-1} e^{-\frac{i}{r}+O(\frac{i^2}{r^2})} \leqslant \frac{1}{2^{k-1}} e^{-\frac{k(k-1)}{2r}+O(\frac{k^3}{r^2})}$. Let $b = \frac{r\ln 2}{p}$, then $b \geqslant r\ln 2 \cdot 2^{k-1} e^{\frac{k(k-1)}{2r}+O(\frac{k^3}{r^2})}$. If we choose $r = \frac{k^2}{2}$ for k sufficiently large, then $b \geqslant \frac{k^2 \ln 2}{2} \cdot 2^{k-1} e^{1+O(1)}$, which implies that we can choose $b = k^2 \cdot 2^k$, and thus get $\Pr[B] < 1$. □

Linearity of Expectation

- $A \subset \Omega$, $X_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$
  $\mathrm{E}[X_A] = \sum_{\omega \in \Omega} \Pr[\omega] X_A(\omega) = \sum_{\omega \in A} \Pr[\omega] = \Pr[A]$
- $X : \Omega \to \mathbb{R}$ is a random variable, $\mathrm{E}[X] = \sum_{\omega \in \Omega} \Pr[\omega] X(\omega)$
- $\Pr[X \geqslant \mathrm{E}[X]] > 0$, $\Pr[X \leqslant \mathrm{E}[X]] > 0$.
- $\mathrm{E}[X+Y] = \mathrm{E}[X] + \mathrm{E}[Y]$

Definition. *Let A be a subset of an additive group. A is sum-free if $\forall\, x, y \in A$, $x + y \notin A$.*

Example.

- $A = \{n+1, n+2, \cdots, 2n\} \subset \mathbb{Z}$
- $A = \{\text{odd integers}\} \subset \mathbb{Z}$

Theorem 15.2. *For any set A of non-zero integers, there is a sum-free set $B \subset A$ with $|B| \geqslant |A|/3$.*

Proof. Let $p = 3k+2$ be a prime *s.t.* $p > 2\max_{a \in A} |a|$. (Such a prime does exist because of Dirichlet's Prime Number Theorem: $\forall\, a, b$ with $(a,b) = 1$, $\exists$ infinitely many primes of the form $a + nb$.)
Let $S = \{k+1, k+2, \cdots 2k+1\} \subset \mathbb{Z}_{3k+2} = \mathbb{Z}_p$, then S is sum-free in $\mathbb{Z}_p$.
For $x \in \mathbb{Z}_p \backslash \{0\}$, define $A_x = \{a \in A : (ax \mod p) \in S\} \subset A$. □

Claim: A_x is sum-free in $\mathbb{Z}$.

Proof. $\forall\ a, b \in A_x$, $(ax \mod p) \in S$ and $(bx \mod p) \in S$. If $a + b \in A_x$, then $((a+b)x \mod p) \in S$, a contradiction to the fact that S is sum-free.
Choose $x \in \mathbb{Z}_p \backslash \{0\}$ uniformly at random, and compute $\mathrm{E}[|A_x|]$.
Define a random variable $X_{a,S} : x \to \begin{cases} 1 & \text{if } (ax \mod p) \in S \\ 0 & \text{otherwise} \end{cases}$. Then $|A_x| = \sum\limits_{a \in A} X_{a,S}$.
So, by linearity of expectation,

$$\mathrm{E}[|A_x|] = \sum_{a \in A} \mathrm{E}[X_{a,S}] = \sum_{a \in A} \Pr[(ax \mod p) \in S]$$

Note that for a fixed $a \in A$, $\{(ax \mod p) : x \in \mathbb{Z}_p \backslash \{0\}\} = \mathbb{Z}_p \backslash \{0\}$, so running over $x \in \mathbb{Z}_p \backslash \{0\}$ for a fixed $a \in A$, there are exactly $|S|$ many $x \in \mathbb{Z}_p \backslash \{0\}$ satisfying $(ax \mod p) \in S$.
So, $E[|A_x|] = \sum\limits_{a \in A} \dfrac{|S|}{p-1} = \sum\limits_{a \in A} \dfrac{k+1}{3k+1} > \sum\limits_{a \in A} \dfrac{1}{3} = \dfrac{|A|}{3}$.
Therefore, $\Pr[|A_x| > \dfrac{|A|}{3}] > 0. \implies \exists\ x \in \mathbb{Z}_p \backslash \{0\}$ *s.t.* A_x is sum-free and $|A_x| \geqslant |A|/3$. $\square$ $\square$

Definition. *A dominating set of $G = (V, E)$ is a set $A \subset V$ s.t. every $v \in V \backslash A$ has a neighbor in A.*

Theorem 15.3. *Let $G = (V, E)$ be a graph on n vertices with minimum degree $\delta > 1$. Then G contains a dominating set of at most $\dfrac{1 + \ln(1+\delta)}{1+\delta} n$ vertices.*

Proof. For $p \in (0, 1)$ which will be determined later, we pick each vertex in V with probability p independently at random.
Let X be the random set of vertices picked. Let Y be the random set of vertices $y \in V \backslash X$ which has no neighbor in X. That is, $y \in Y$ iff y is not picked and all neighbors of y are not picked. We consider $X \cup Y$.
$\mathrm{E}[|X \cup Y|] = \mathrm{E}[|X|] + \mathrm{E}[|Y|]$.
$\mathrm{E}[|X|] = \sum\limits_{x \in V} \Pr[x \in X] = np$.
$\Pr[y \in Y] = (1-p) \cdot (1-p)^{deg(y)} \leqslant (1-p)^{1+\delta}$.
$\implies \mathrm{E}[|Y|] = \sum\limits_{y \in V} \Pr[y \in Y] \leqslant n(1-p)^{1+\delta}$.
So, $\mathrm{E}[|X \cup Y|] \leqslant np + n(1-p)^{1+\delta} \leqslant np + ne^{-p(1+\delta)}$.
Choose $p = \dfrac{\ln(1+\delta)}{1+\delta}$ such that $p + e^{-p(1+\delta)}$ is minimized, then $\mathrm{E}[|X \cup Y|] \leqslant \dfrac{1 + \ln(1+\delta)}{1+\delta} n$.
$\implies \exists$ a dominating set of size $\dfrac{1 + \ln(1+\delta)}{1+\delta} n$. $\square$

16 Linearity of Expectation, Deletion Method

Theorem 16.1. *$G = (V, E)$, $|V| = n$, $|E| = m$, then G contains a bipartite subgraph with at least $\frac{m}{2}$ edges.*

Proof. Consider a random partition $L \cup R = V$. $\forall v \in V$, put it into L or R with equal probability, independently. Let X be the number of crossing edges from L to R. Let X_{uv} be the indicator random variable of the event that the edge uv is crossing, then $X = \sum_{u \sim v} X_{uv}$

E$[X_{uv}]$ =Pr$[uv$ is crossing$]$=Pr$[u \in L, v \in R$ or $u \in R, v \in L]$=$\frac{1}{2}$
E$[X]$= $\sum_{u \sim v}$ E$[X_{uv}]$=$\frac{m}{2}$
$\Longrightarrow \exists$ a bipartite subgraph with $\geq \frac{m}{2}$ edges. □

Recall: $\alpha(G)$: independent number

- $\chi(G)\alpha(G) \geq |G|$
- $\alpha(G) \leq n \frac{-\lambda_n}{\lambda_1 - \lambda_n}$

Theorem 16.2. *For any graph G, $\alpha(G) \geq \sum_{v \in V} \dfrac{1}{1 + deg(v)}$.*

Proof. Let $V(G) = [n]$. For $i \in [n]$, let $N_i = \{j \in [n] : j \sim i\}$, let S_n be the symmetric group over $[n]$.
For given $\pi \in S_n$ we say a vertex $i \in [n]$ is $\pi-$dominating, if $\pi(i) < \pi(j)$ for all $j \in N_i$. Let M_π ={all $\pi-$dominating vertices.}

<u>**Claim:**</u> $\forall \pi \in S_n$, M_π is a independent set.
<u>**proof of claim:**</u> Suppose not, then $\exists i, j \in M_\pi$ with $i \sim j$. $i \in N(j) \Rightarrow \pi(j) < \pi(i)$ and $j \in N(i) \Rightarrow \pi(i) < \pi(j)$, contradiction!

Pick $\pi \in S_n$ uniformly at random, compute $E[|M_\pi|]$.
For any fixed $i \in [n]$, let $I_{\{i \text{ is } \pi-\text{dominating}\}}$ be a indicator random variable, then $|M_\pi| = \sum_{i \in [n]} I_{\{i \text{ is } \pi-\text{dominating}\}}$

$$\begin{aligned}
E[|M_\pi|] &= \sum_{i \in [n]} E[I_{\{i \text{ is } \pi-\text{dominating}\}}] \\
&= \sum_{i \in [n]} Pr[i \text{ is } \pi - \text{dominating}] \\
&= \sum_{i \in [n]} \frac{\binom{n}{deg(i)+1} \cdot deg(i)! \cdot (n - deg(i) - 1)!}{n!} \\
&= \sum_{i \in [n]} \frac{1}{deg(i) + 1}.
\end{aligned}$$

$\Longrightarrow \exists$ independent set of size $\geq \sum_{i\in[n]} \frac{1}{deg(i)+1}$

$\square$

Corollary 16.3. *$\forall G$ with m edges and n vertices, then $\alpha(G) \geq \dfrac{n^2}{2m+n}$*
$m = \dfrac{nd}{2}$ where d is average degree, then $\alpha(G) \geq \dfrac{n}{1+d}$.

Proof. $2m = \sum_{v\in V} deg(v)$, then $\sum_{v\in V}(deg(v)+1) = 2m+n$.
By Cauchy-Schwarz Inequality, $\alpha(G) \geq \sum_{v\in V} \frac{1}{1+deg(v)} \geq \frac{n^2}{\sum_{v\in V}(1+deg(v))} = \frac{n^2}{2m+n}$

$\square$

The deletion method

• **Ideas:** A random structure doesn't always have the directed property, and may have some very few "blemishes". After deleting all blemishes, we will obtain the wanted structure.

Theorem 16.4. *Let G be a graph on n vertices and with average degree d, then $\alpha(G) \geq \dfrac{n}{2d}$*

Proof. Let $S \subset V(G)$ be a random set, for $\forall v \in V, Pr(v \in S) = p$ and value of p will be determined later. Let $X = |S|$ and $Y = \#$edges in S. Then $E[X] = np, E[Y] = |E(G)| \cdot p^2 = \dfrac{nd}{2}p^2$. Then $E[X-Y] = np - \dfrac{nd}{2}p^2 = n(p - \dfrac{d}{2}p^2)$.
By choosing $p = \dfrac{1}{d}$, we have $E[X-Y] = \dfrac{n}{2d}$ which is maximum.
So there is a particular set S such that $|S| - Y \geq E[X-Y] = \dfrac{n}{2d}$.
Now deleting one vertex form each edge of S, leaving a set S'. This set S' is independent and has at least $\dfrac{n}{2d}$ vertices. $\square$

Recall: $R(k,k) > \dfrac{1}{e\sqrt{2}}k2^{k/2}$.

Theorem 16.5. $\forall n,\ R(k,k) > n - \dbinom{n}{k}2^{1-\binom{k}{2}}$

Proof. Consider a random 2−edge-coloring of K_n, where each edge is colored by red or blue with probability $\dfrac{1}{2}$, independently. For $A \in \dbinom{[n]}{k}$, Let X_A be the indicator random variable of the event that A is monochromatic. Let X be the number of monochromatic k cliques.
$E[X] = \sum_{A\in\binom{[n]}{k}} E[X_A] = \dbinom{n}{k}2^{1-\binom{k}{2}}$.

Then there exists a $2-$edge-coloring of K_n, s.t. there are $\leq E[X] = \binom{n}{k} 2^{1-\binom{k}{2}}$ monochromatic k cliques. Fix such a coloring, remove one vertex from each monochromatic $k-$subset. This will delete $\leq \binom{n}{k} 2^{1-\binom{k}{2}}$ vertices, which has No monochromatic K_k. So $R(k,k) > n - \binom{n}{k} 2^{1-\binom{k}{2}}$. □

Corollary 16.6. $R(k,k) > \frac{1}{e}(1+O(1))k2^{k/2}$.

Definition. *A clique covering of a graph G is a set $H_1, \cdots, H_t$ of its clique subgraphs such that each edge of G belongs to at least one of those cliques. Denote $cc(G)$ = min $\sharp$ cliques in clique covering of G.*

Theorem 16.7. *Let G be a graph on n vertices such that every vertex has at least one neighbor and at most d non-neighbors. Then $cc(G) = O(d^2 \ln n)$.*

Proof. To be continued. □

17 Deletion Method, Markov's Inequality, Random Graph

Recall. $cc(G) = \min \#$ cliques in a clique covering of G

Theorem 17.1. *Let G be a graph on n vertices s.t. every vertex has at least one neighbor and at most d non-neighbors. Then*

$$cc(G) \leqslant O(d^2 \ln n).$$

Proof. Consider the following way of choosing cliques of $G = (V, E)$.
First, pick $v \in V$ independently with probability $p = \frac{1}{1+d}$ to get a set $W \subset V$. Then remove from W all vertices having at least one non-neighbor in W. Then we get a clique of G. Apply this way independently t times to get t cliques $H_1, \cdots, H_t$.
Let X be the number of edges not covered by any H_i. Let X_{uw} be the indicator random variable of the event that $u \sim w$ is not covered by any H_i. Then $X = \sum\limits_{u \sim w} X_{uw}$.
Note that H_i covers $u \sim w$ if both u and w are chosen and none of their $\leqslant 2d$ non-neighbors are chosen. So

$$\begin{aligned}\Pr[u \sim w \text{ is covered by } H_i] &\geqslant p^2(1-p)^{2d} = \frac{1}{(1+d)^2}(\frac{d}{1+d})^{2d} \\ &= \frac{1}{(1+d)^2}\frac{1}{(1+\frac{1}{d})^{2d}} \geqslant \frac{1}{(1+d)^2 e^2}\end{aligned}$$

So

$$\begin{aligned}\Pr[u \sim w \text{ is not covered by any } H_i] &= \Pr[\bigcap_{i=1}^{t} \{u \sim w \text{ is not covered by } H_i\}] \\ &= \prod_{i=1}^{t} \Pr[u \sim w \text{ is not covered by } H_i] \\ &\leqslant (1 - \frac{1}{(1+d)^2 e^2})^t \leqslant e^{-\frac{t}{(1+d)^2 e^2}}\end{aligned}$$

Then, by linearity of expectation,

$$\begin{aligned}\mathrm{E}[X] &= \sum_{u \sim w} \mathrm{E}[X_{uw}] = \sum_{u \sim w} \Pr[u \sim w \text{ is not covered by any } H_i] \\ &\leqslant \binom{n}{2} e^{-\frac{t}{(1+d)^2 e^2}} \leqslant \frac{e^{2 \ln n}}{2} e^{-\frac{t}{(1+d)^2 e^2}}\end{aligned}$$

Choose $t \geqslant 2 \ln n (1+d)^2 e^2$, $t = O(d^2 \ln n)$, we have $\mathrm{E}[X] < 1$. Hence, there is at least one choice of $t = O(d^2 \ln n)$ cliques that form a clique covering of G. $\implies cc(G) \leqslant O(d^2 \ln n)$. □

Markov's Inequality. Let $X \geqslant 0$ be a random variable, $a > 0$, then $\Pr[X \geqslant a] \leqslant \dfrac{\mathrm{E}[X]}{a}$.

Proof. Let $I_{\{X \geqslant a\}}$ be the indicator random variable.
Then $aI_{\{X \geqslant a\}} \leqslant X$. $\implies \mathrm{E}[aI_{\{X \geqslant a\}}] = a\Pr[X \geqslant a] \leqslant \mathrm{E}[X]$. □

Corollary 17.2. *Let $X_n \geqslant 0$ be integer-valued random variables in (Ω_n, Pr_n), $n \in \mathbb{Z}_{>0}$. If $E[X_n] \to 0$ as $n \to \infty$, then $Pr[X_n = 0] \to 1$ as $n \to \infty$.*

Proof. $\Pr[X_n \geqslant 1] \leqslant \mathrm{E}[X_n] \to 0$ as $n \to \infty$. □

Definition. *Let $X_n \geqslant 0$ be integer-valued random variables in (Ω_n, Pr_n), $n \in \mathbb{Z}_{>0}$. If $Pr[X_n = 0] \to 1$ as $n \to \infty$, we say $X_n = 0$ almost surely occur.*

Definition. *The random graph $G(n,p)$ for $0 \leqslant p \leqslant 1$ is a graph with vertex set $[n]$, where each of the potential $\binom{n}{2}$ edges appears with probability p independently.*

Theorem 17.3. *$G(n, \frac{1}{2})$ almost surely is NOT bipartite.*

Proof. Let A_n be the event that $G(n, \frac{1}{2})$ is bipartite. Then it suffices to show $\Pr[A_n] \to 0$ as $n \to \infty$.
For $U \in 2^{[n]}$, let A_U be the event that all edges of G are between U and $[n] \backslash U$.

Then $A_n = \cup_{U \in 2^{[n]}} A_U$.

$$\Pr[A_U] = (\frac{1}{2})^{\binom{|U|}{2}} (\frac{1}{2})^{\binom{n-|U|}{2}} = \frac{1}{2^{\binom{|U|}{2}+\binom{n-|U|}{2}}} \leqslant \frac{1}{2^{2\binom{n/2}{2}}} = 2^{-\frac{n^2}{4}+\frac{n}{2}}$$

So

$$\Pr[A_n] \leqslant 2^n \cdot 2^{-\frac{n^2}{4}+\frac{n}{2}} = 2^{-\frac{n^2}{4}+\frac{3n}{2}} \to 0 \text{ as } n \to \infty$$

□

Theorem 17.4. *$G(n,p)$ for fixed $p \in (0,1)$, then*

$$Pr[\alpha(G) \leqslant \left\lceil \frac{2\ln n}{p} \right\rceil] \to 1 \text{ as } n \to \infty.$$

Proof. Let $k = \left\lceil \frac{2\ln n}{p} \right\rceil$. Let X_n be the number of independent sets of size $k+1$. For $\forall\ S \in \binom{[n]}{k+1}$, let $I_{\{S \text{ is independent}\}}$ be the indicator random variable, then $X_n = \sum\limits_{S \in \binom{[n]}{k+1}} I_{\{S \text{ is independent}\}}$.

$$\mathrm{E}[I_{\{S \text{ is independent}\}}] = \Pr[S \text{ is independent}] = (1-p)^{\binom{k+1}{2}}$$

So,

$$\begin{aligned}\mathrm{E}[X_n] &= \sum_{S \in \binom{[n]}{k+1}} \mathrm{E}[I_{\{S \text{ is independent}\}}] = \binom{n}{k+1}(1-p)^{\binom{k+1}{2}} \\ &\leqslant \frac{n^{k+1}}{(k+1)!} e^{-p\binom{k+1}{2}} = \frac{(ne^{-p\frac{k}{2}})^{k+1}}{(k+1)!} < \frac{1}{(k+1)!} \to 0 \text{ as } n \to \infty\end{aligned}$$

By Corollary, $\Pr[X_n = 0] \to 1$ as $n \to \infty$.
$\Longrightarrow \Pr[\alpha(G) \leqslant k] \to 1$ as $n \to \infty$. □

Definition. *The girth of G, denoted by $g(G)$, is the length of the shortest cycle in G.*

Theorem 17.5. *For $\forall\ k \in \mathbb{N}^+$, there exists a graph G with $\chi(G) \geqslant k$ and $g(G) \geqslant k$.*

Proof. Consider $G = G(n,p)$ where p will be determined later.
Let X be the number of cycles of length less than k. Let X_i be the number of cycles of length i. Then $X = \sum\limits_{i=3}^{k-1} X_i$.

$$\begin{aligned}\mathrm{E}[X_i] &= \sum_{(x_1 \cdots x_i)} \mathrm{E}[I_{\{(x_1 \cdots x_i) \text{ is a cycle}\}}] = \sum_{(x_1 \cdots x_i)} \Pr[(x_1 \cdots x_i) \text{ is a cycle}] \\ &= \frac{n(n-1)\cdots(n-i+1)}{2i} p^i\end{aligned}$$

So

$$\mathrm{E}[X] = \sum_{i=3}^{k-1} \frac{n(n-1)\cdots(n-i+1)}{2i} p^i \leqslant \sum_{i=0}^{k-1} (np)^i = \frac{(np)^k - 1}{np - 1}.$$

By Markov's Inequality,

$$\Pr[X \geqslant \frac{n}{2}] \leqslant \frac{\mathrm{E}[X]}{\frac{n}{2}} \leqslant \frac{2[(np)^k - 1]}{n(np-1)}.$$

Let $p = n^{-\frac{k-1}{k}}$, then $\Pr[X \geqslant \frac{n}{2}] \leqslant \frac{2(n-1)}{n(n^{\frac{1}{k}} - 1)} \to 0$ as $n \to \infty$.

Delete a vertex from each cycle of length less than k, we have a graph $G' \subset G$ with no cycle of length less than k.

Note that $|G'| \geqslant n - \frac{n}{2} = \frac{n}{2}$, $g(G') \geqslant k$.

Recall that $\chi(G) \geqslant \frac{|G|}{\alpha(G)}$, and $\Pr[\alpha(G) \leqslant \left\lceil \frac{2\ln n}{p} \right\rceil] \to 1$ as $n \to \infty$.

So $\alpha(G') \leqslant \alpha(G) \leqslant \left\lceil \frac{2\ln n}{p} \right\rceil \leqslant 3\ln n \cdot n^{\frac{k-1}{k}}$.

$\Longrightarrow \chi(G') \geqslant \frac{|G'|}{\alpha(G')} \geqslant \frac{\frac{n}{2}}{3\ln n \cdot n^{\frac{k-1}{k}}} = \frac{n^{\frac{1}{k}}}{6\ln n} \gg k$ as $n \to \infty$. $\square$